The Illusion of Learning from Observational Data: An Empirical Bayes Perspective

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Abstract: Randomized experiments have long been the gold standard for scientists seeking to learn about cause and effect. When randomized experiments are infeasible, scientists often resort to observational studies for causal inference. Observational studies are widely available and often large, but they rely on untestable assumptions that, when violated, may result in biased estimates. Uncertainty about bias leads to a phenomenon known as the *illusion of learning from observational research* (Gerber, Green and Kaplan, 2004): absent prior information about bias, observational results cannot meaningfully contribute to the estimation of a causal parameter. To shatter the illusion, we take an empirical Bayes perspective. We show that the distribution of observational biases can be learned from *calibration studies*—experiments that target a causal effect that is known *a priori* to be zero. Calibration identifies the distribution of observational bias and allows observational studies to inform the estimation of causal parameters via empirical Bayes shrinkage. We formalize the illusion phenomenon in an empirical Bayes setting and show that, with an increasing number of calibration and observation studies, both the bias distribution and the causal effect can be consistently recovered. We illustrate our method through a simulation study and a semi-synthetic application based on Ferraro and Miranda (2013)'s water-usage experiment.

Keywords and phrases: causal inference, empirical Bayes, observational studies, calibration studies.

1. Introduction

Randomized experiments are the gold standard for estimating causal effects. By design, they break the link between treatment assignment and confounding factors, so they identify causal effects under relatively minimal assumptions (Imbens and Rubin, 2015; Ding, 2024). But experiments are often expensive, difficult, or unethical to run at scale.

Observational studies offer the opposite trade-off. They are usually plentiful, cheap, and ethnically unencumbered. But because treatment is not allocated

randomly, observational studies may be biased. This raises a basic question: when both experimental and observational evidence are available, can we combine them to learn more about a causal effect than we would learn from the experiment alone?

Gerber, Green and Kaplan (2004) makes a Bayesian argument to show that the answer is not always yes. It presents a Bayesian model of experimental and observational data, in which the observational estimates may be biased by an unknown amount. When the researcher is completely uncertain about that bias, i.e., when the prior is flat, the observational studies receive no weight in the posterior for the target causal effect. In this sense, the observational studies create only an *illusion of learning*: they appear relevant but they do not improve inference about the causal quantity of interest.

This result clarifies the real obstacle. The problem is not observational data per se. Rather, it is that we do not know how their biases are distributed. If we knew something about the population distribution of biases induced by a given observational design—that is, if we held an informative prior—then observational studies could contribute to the estimate of the causal effect.

Our idea is that this perspective suggests an empirical Bayes approach. When we have many observational studies, we can view their study-specific biases as draws from a population, and we can try to learn that population from data. Empirical Bayes is a natural framework for this problem (Robbins, 1956; Efron, 2012; Ignatiadis and Sen, 2025): it uses repeated noisy estimates to fit a prior distribution, and then uses that fitted prior to improve estimation. In our setting, the repeated estimates are observational studies, and the unknown prior is the distribution of their biases.

Thus we study when and how empirical Bayes can overcome the illusion of learning from observational data. First, we show that if we assume the biases have mean zero and estimate only their variance, then observational studies can improve inference, thereby "breaking" the illusion. The posterior variance of the estimate goes down, and the risk vanishes as we see more observational studies.

But on further reflection, the zero-mean assumption is not credible. It presumes that, on average, the bias of all observational studies cancels out. To relax this assumption, we then consider analyzing the observational studies to estimate both the mean and variance of the prior on the biases. With this procedure, however, we show that the illusion returns. The observational studies do not move the posterior mean of the causal effect away from the experimental estimate. Even worse, they make the posterior variance smaller, creating unwarranted confidence in the experimental results.

Finally, we show how to break the illusion without relying on the unrealistic assumption that observational biases average to zero. The idea is to conduct *calibration studies*: observational studies of interventions whose true causal effect is known in advance, typically to be zero. Because their true effect is known, any nonzero estimate from a calibration study can be attributed to its bias.

Calibration studies help solve our problem because they provide direct information about the distribution of biases associated with a given observational design. This suggests that we use empirical Bayes on the calibration studies

to fit the prior over the biases and then use the fitted prior to combine the experimental estimate and a collection of observational estimates. The result is a *calibrated empirical Bayes* procedure for causal estimation from experimental and observational studies.

In sum, our paper makes the following contributions. First, we formalize the illusion of learning from observational data in a hierarchical Gaussian model for one experimental estimate and many observational estimates. Second, we show that applying empirical Bayes to the variance of the biases successfully breaks the illusion but requires the unrealistic assumption that we know their mean. Third, we show that applying empirical Bayes to both the mean and variance of the biases brings back the illusion, where the observational studies do not inform the estimate. Finally, we show that calibration studies restore the possibility of learning: they identify the bias distribution such that observational studies can improve estimation of the causal effect. With calibrated empirical Bayes, the causal estimator achieves vanishing risk as the number of observational and calibration studies grows.

We study this method in theory, simulation, and a semi-synthetic analysis based on a large field experiment. Calibrated empirical Bayes improves precision relative to the experiment alone, enabling observational studies to contribute meaningfully to the estimation of a causal effect.

2. Related work

Our paper connects to three strands of work. First, there is a large literature comparing experimental and observational studies. Some meta-epidemiological and benchmarking studies find that well-designed observational studies can track experimental results, while others document substantial discrepancies (Benson and Hartz, 2000; Concato, Shah and Horwitz, 2000; Anglemyer, Horvath and Bero, 2014; LaLonde, 1986; Dehejia and Wahba, 1999; Smith and Todd, 2001; Wong, Steiner and Anglin, 2018; Bloom, 2005; Gordon et al., 2019; Arceneaux, Gerber and Green, 2010; Waddington, Villar and Valentine, 2023). This literature shows both the promise and the risk of observational evidence, but it does not provide a general method for learning the bias distribution of a new collection of observational studies.

Second, our work relates to methods that combine experimental and observational studies. Existing approaches use transportability assumptions, bias adjustment, shrinkage, reweighting, or surrogate outcomes to borrow strength across data sources (Cole and Stuart, 2010; Stuart, 2010; Guo et al., 2021; Turner et al., 2009; Athey, Imbens and Wager, 2018; Yang et al., 2023; Imbens et al., 2025; Rosenman et al., 2023; Xiong et al., 2023; Yang et al., 2025; Lin, Bickel and Ding, 2026; Yang and Ding, 2020). These methods are valuable, but they require structure on how observational and experimental evidence are related. Our focus is different: we ask what can be learned when the main obstacle is uncertainty about the distribution of observational bias itself.

Third, our calibration studies are close in spirit to negative controls and related

diagnostic tools (Lipsitch, Tchetgen Tchetgen and Cohen, 2010; Tchetgen Tchetgen, 2014; Miao, Geng and Tchetgen Tchetgen, 2018). Like that literature and related placebo-controlled experiments that address noncompliance, we use interventions with known null effects to reveal bias. Our contribution is to place that idea inside an empirical Bayes framework for combining one experiment with many observational studies. The key object is not a single bias diagnostic, but an estimated population distribution of study-specific biases.

3. A Bayesian approach to combining experimental and observational data

Suppose we run an experiment to estimate a true causal effect $\theta^\star$,

$$y_{\mathrm{e}} \sim \mathcal{N}(\theta^\star, \sigma_{\mathrm{e}}^2). \tag{3.1}$$

Here σ_{e}^2 is the known variance, based on the size of the experiment.

Suppose we also collect observational studies $\mathbf{y}_{\mathrm{o}} = \{y_{\mathrm{o},j}\}_{j=1}^J$, each of which provides a biased estimate of the same causal effect. The bias of study j is denoted b_j^*,

$$y_{\mathrm{o},j} \sim \mathcal{N}(\theta^\star + b_j^\star, \sigma_{\mathrm{o},j}^2). \tag{3.2}$$

Here $\sigma_{\mathrm{o},j}^2$ is again a known variance, based on the size of the study. The normal means model is a reasonable theoretical device: the sampling distribution of causal estimators is often asymptotically normal; see, for example, results such as Theorem 4.2 of Ding (2024).

Our goal is to combine the experimental and observational studies to estimate $\theta^\star$. We will take a Bayesian perspective. We form a hierarchical model of the experiment and observational studies:

Experimental study:

$$\theta \sim \mathrm{p}(\theta) \propto 1 \tag{3.3}$$

$$y_{\mathrm{e}} \sim \mathcal{N}(\theta, \sigma_{\mathrm{e}}^2) \tag{3.4}$$

Observational studies $j \in [J]$:

$$b_j \sim \mathrm{p}(b) \tag{3.5}$$

$$y_{\mathrm{o},j} \sim \mathcal{N}(\theta + b_j, \sigma_{\mathrm{o},j}^2). \tag{3.6}$$

With this model, we estimate the causal effect through its posterior $\mathrm{p}(\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}})$. The posterior mean $\hat{\theta}$ provides a point estimate; the posterior variance $\hat{v}$ provides (model-based) uncertainty. We might calculate the posterior exactly or approximately. Below, we will also consider the risk of different estimates relative to the true $\theta^\star$

In this model, note the flat (improper) prior on θ, which reflects that we have no prior information about the true effect. But we have not yet specified the prior of the bias $\mathrm{p}(b)$. Below, we will consider different ways to set that prior,

including uninformative priors and empirical Bayes. For each, we will consider the posterior mean and variance of θ, as well as properties of the risk of the posterior mean $\hat{\theta}$ relative to $\theta^\star$,

$$R(\hat{\theta}, \theta^\star) = \mathbb{E}\left[(\hat{\theta} - \theta^\star)^2\right]. \tag{3.7}$$

Here the expectation is with respect to the data that forms the estimate $\hat{\theta}$.

3.1. A model of experimental data alone

We first consider our model without any observational data. With the flat prior, the posterior mean and variance are

$$\hat{\theta}_{\mathrm{e}} := \mathbb{E}\left[\theta \mid y_{\mathrm{e}}\right] = y_{\mathrm{e}} \tag{3.8}$$

$$\hat{v}_{\mathrm{e}} := \mathrm{Var}\left[\theta \mid y_{\mathrm{e}}\right] = \sigma_{\mathrm{e}}^2. \tag{3.9}$$

This estimator is unbiased for $\theta^\star$ and its risk is $R(\hat{\theta}_{\mathrm{e}}, \theta^\star) = \sigma_e^2$.

The experimental estimator $\hat{\theta}_e$ often serves as the gold standard for causal inference. The difficulty is that experiments are often limited by cost, ethics, or feasibility, so σ_e^2 is often large. This issue motivates researchers to try to supplement experimental evidence with observational studies.

3.2. A flat prior on the biases, and the illusion of learning from observational data

We might be tempted to set $\mathrm{p}(b)$ in Eq. (3.5) to be a flat prior as well, reflecting that we have no prior information about the biases of the observational studies. But this choice results in the *illusion of learning from observational data* (Gerber, Green and Kaplan, 2004). As the next theorem shows, with flat priors on both the effect and the biases, $\mathbb{E}\left[\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}\right] = y_{\mathrm{e}}$ and the posterior variance remains the same, $\mathrm{Var}\left[\theta \mid y_{\mathrm{e}}\right] = \sigma_{\mathrm{e}}^2$. The observational studies offer no information beyond the experiment.

Theorem 1 (The illusion of learning from observational data)**.** *We observe an experimental study and J observational studies $\{y_{\mathrm{e}}, y_{\mathrm{o},1:J}\}$. We consider the model of Eqs. (3.3) to (3.6) with a flat prior for $\mathrm{p}(b)$.*

The posterior $\mathrm{p}(\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}})$ is a normal distribution,

$$\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \sim \mathcal{N}(\hat{\theta}_{\mathrm{flat}}, \hat{v}_{\mathrm{flat}}^2), \tag{3.10}$$

The posterior mean is

$$\hat{\theta}_{\mathrm{flat}} = \mathbb{E}_{\mathrm{flat}}\left[\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}\right] = y_e \tag{3.11}$$

The posterior variance is

$$\hat{v}_{\mathrm{flat}} = \mathrm{Var}_{\mathrm{flat}}\left[\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}\right] = \sigma_e^2. \tag{3.12}$$

The posterior mean is an unbiased estimate with risk $R(\hat{\theta}_{\mathrm{flat}}, \theta^\star) = \sigma_{\mathrm{e}}^2$.

Theorem 1 shows that the observational data plays no role in the estimate, which is equivalent to having collected the experimental data alone. It is a special case of the result in Gerber, Green and Kaplan (2004).

3.3. Empirical Bayes (of the variance) breaks the illusion

The hierarchical model and Theorem 1 assume multiple observational studies. This suggests an opportunity to use *empirical Bayes* (EB) to set the prior (Robbins, 1956; Efron, 2012; Ignatiadis and Sen, 2025). The general idea behind EB is that multiple estimates provide information about an unknown prior, which is fit so that the model matches the population, e.g., by maximum likelihood or moment matching. EB appeals to both Bayesian and frequentist perspectives on statistics in that a Bayesian model is fit to have good frequentist properties.

Suppose now the prior on the biases is a zero-mean normal with an unknown variance $b_j \sim \mathcal{N}(0, \gamma^2)$. Following one approach to EB, we can fit the variance to match the population by maximizing the marginal likelihood,

$$\hat{\gamma}^2 \in \arg\max_{\gamma^2 \geq 0} \log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \,;\, \gamma^2), \tag{3.13}$$

where the marginal likelihood is

$$\log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \,;\, \gamma^2) = \int \mathrm{p}(\theta)\, \mathrm{p}(y_{\mathrm{e}} \,|\, \theta) \left(\prod_{j=1}^{J} \mathrm{p}(b_j \,;\, \gamma^2)\, \mathrm{p}(y_{\mathrm{o},j} \,|\, \theta, b_j) \right) \mathrm{d}\theta. \tag{3.14}$$

We then substitute $\hat{\gamma}^2$ into the model and calculate the posterior estimate of θ. See the left panel of Figure 3.1 for the graphical model.

The next theorem shows that this form of EB breaks the illusion of learning from observational studies. The observational studies now adjust the posterior expectation of the effect and reduces the posterior variance. The risk goes to zero as we see more observational studies.

Theorem 2 (No illusion with zero-mean bias)**.** *We observe an experimental study and J observational studies $\{y_{\mathrm{e}}, y_{\mathrm{o},1:J}\}$, where $J > 1$. Assume $b \sim \mathcal{N}(0, \gamma^2)$ and fit $\hat{\gamma}^2$ from Eq. (3.13).*

The posterior $\mathrm{p}(\theta \,|\, y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \,;\, \hat{\gamma}^2)$ is a normal distribution,

$$\theta \,|\, y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \sim \mathcal{N}(\hat{\theta}_{\mathrm{EB}_0}, \hat{v}^2_{\mathrm{EB}_0}). \tag{3.15}$$

The posterior mean is

$$\hat{\theta}_{\mathrm{EB}_0} = y_{\mathrm{e}} + \frac{\sum_{j=1}^{J} \left(\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2\right)^{-1} (y_{\mathrm{o},j} - y_{\mathrm{e}})}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} \left(\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2\right)^{-1}}. \tag{3.16}$$

The posterior variance is

$$\hat{v}^2_{\mathrm{EB}_0} = \left(\sigma_{\mathrm{e}}^{-2} + \textstyle\sum_{j=1}^{J} \left(\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2\right)^{-1}\right)^{-1}. \tag{3.17}$$

Under the assumption that there exists a true prior $\gamma^{2\star}$, the estimator $\hat{\theta}_{\mathrm{EB}_0}$ achieves asymptotically vanishing risk as we see more observational studies,

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) \to 0 \quad \text{as } J \to \infty. \tag{3.18}$$

(See Theorem 5 in the Appendix for results about the risk and consistency of the EB procedure.)

Eqs. (3.16) and (3.17) come from standard calculations for hierarchical Gaussian models; see Proposition 1 in the appendix. The theorem shows that by assuming the biases have zero mean and an unknown variance, we can extract signal from observational studies. The posterior mean in Eq. (3.16) is shrunk towards the observational studies; the posterior variance in Eq. (3.17) goes down. Relative to the true $\theta^\star$, the error in our estimate becomes smaller as we see more observational studies. The proof is in the appendix; it uses sample splitting to show the results.

This assumption is strong. It says that, although individual observational studies may be biased, those biases average to zero across studies.

3.4. Empirical Bayes of the mean and variance, and the return of the illusion

It might seem we have broken the illusion. But the assumption that $b \sim \mathcal{N}(0, \gamma^2)$ is strong, even when we estimate γ^2. This prior assumes that while observational studies are biased, the biases themselves have mean zero. There is usually no reason for this assumption to be true, though we will return to it in the discussion where we connect it to arguments for multi-method and triangulation-based research in the social sciences (Brewer and Hunter, 2006; Brady and Collier, 2010; Creswell and Creswell, 2018).

One way to relax the strong assumption could be to give the observational studies an unknown mean as well. We can assume that $b \sim \mathcal{N}(\mu, \gamma^2)$, and fit both the mean and variance with EB:

$$\hat{\mu}, \hat{\gamma}^2 \in \underset{\mu \in \mathbb{R}, \gamma^2 \geq 0}{\arg\max} \log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}\,;\, \mu, \gamma^2). \tag{3.19}$$

The marginal likelihood is similar to Eq. (3.14), but also involves the mean,

$$\log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}\,;\, \mu, \gamma^2) = \int \mathrm{p}(\theta)\, \mathrm{p}(y_{\mathrm{e}} \,|\, \theta) \left(\prod_{j=1}^{J} \mathrm{p}(b_j\,;\, \mu, \gamma^2)\, \mathrm{p}(y_{\mathrm{o},j} \,|\, \theta, b_j) \right) \mathrm{d}\theta. \tag{3.20}$$

We substitute $\hat{\mu}, \hat{\gamma}^2$ into the model and calculate the posterior estimate of θ. See the middle panel of Figure 3.1 for the graphical model.

But the next theorem shows that when we fit both the mean and variance of the biases, the illusion returns. Even worse, it shows that the posterior variance

of the estimated effect goes *down*. The observational studies, which do not inform our estimate of the effect, only make us overconfident about the results of the experiment.

Theorem 3 (The illusion returns)**.** *We observe an experimental study and J observational studies $\{y_{\mathrm{e}}, y_{\mathrm{o},1:J}\}$. Assume $b_j \sim \mathcal{N}(\mu, \gamma^2)$ and fit $\hat{\mu}, \hat{\gamma}^2$ from Eq. (3.19).*

The posterior $\mathrm{p}(\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}\,;\, \hat{\mu}, \hat{\gamma}^2)$ is a normal distribution,

$$\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \sim \mathcal{N}(\hat{\theta}_{\mathrm{EB}}, \hat{v}^2_{\mathrm{EB}}). \tag{3.21}$$

The posterior mean is

$$\hat{\theta}_{\mathrm{EB}} = y_{\mathrm{e}}. \tag{3.22}$$

The posterior variance is

$$\hat{v}^2_{\mathrm{EB}} = \left(\sigma_{\mathrm{e}}^{-2} + \textstyle\sum_{j=1}^{J} \left(\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2\right)^{-1}\right)^{-1}. \tag{3.23}$$

The risk is equal to the variance of the experiment $R(\hat{\theta}_{\mathrm{EB}}, \theta^\star) = \sigma^2_{\mathrm{e}}$

Again, Eq. (3.22) shows that our inference of the effect θ does not include any information from the observational studies. The risk is the same as if we saw experiment alone. Paradoxically, even though they do not inform the estimate, Eq. (3.23) shows that our model-based confidence in the estimate goes up as we see more studies.

3.5. Using calibration studies to break the illusion

How can we break the illusion without making an unrealistic assumption about the distribution of biases? To solve this problem, we collect *calibration data* or *negative controls*. These are observational studies that target a known zero causal effect, and for which their biases come from the same distribution of biases as our original observational studies.

We now include these studies in our hierarchical model:

Experimental study:

$$\theta \sim \mathrm{p}(\theta) \propto 1, \tag{3.24}$$

$$y_{\mathrm{e}} \sim \mathcal{N}(\theta, \sigma^2_{\mathrm{e}}). \tag{3.25}$$

Observational studies $j \in [J]$:

$$b_{\mathrm{o},j} \sim \mathcal{N}(\mu, \gamma^2), \tag{3.26}$$

$$y_{\mathrm{o},j} \sim \mathcal{N}(\theta + b_{\mathrm{o},j}, \sigma^2_{\mathrm{o},j}). \tag{3.27}$$

Calibration studies $k \in [K]$:

$$b_{\mathrm{c},k} \sim \mathcal{N}(\mu, \gamma^2), \tag{3.28}$$

$$y_{\mathrm{c},k} \sim \mathcal{N}(b_{\mathrm{c},k}, \sigma^2_{\mathrm{c},k}). \tag{3.29}$$

This is a model of experimental data, observational studies, and calibration studies. Eq. (3.29) assumes that each calibration study $y_{\mathrm{c},k}$ is an unbiased estimate of the per-study bias $b_{\mathrm{c},k}$. It is unbiased because this likelihood does not involve the effect θ, unlike the likelihood for the observational studies in Eq. (3.27). Crucially, the priors of the per-study biases, for both observational and calibration studies, come from the same distribution; see Eq. (3.28) and Eq. (3.26). Note the model does *not* assume that the calibration studies are paired with the observational studies, only that their per-study biases come from the same population.[1]

To form our estimate of the effect, we first fit the mean μ and variance γ^2 by maximizing the marginal likelihood of the calibration studies:

$$\hat{\mu}, \hat{\gamma}^2 \in \underset{\mu \in \mathbb{R}, \gamma^2 \geq 0}{\arg\max} \log \mathrm{p}(\mathbf{y}_{\mathrm{c}} \,;\, \mu, \gamma^2). \tag{3.30}$$

We then substitute $\hat{\mu}, \hat{\gamma}^2$ into the full model and calculate the posterior of θ. See the right panel of Figure 3.1 for the graphical model. The next theorem shows that with calibration studies, the observational data meaningfully influences our estimate of the causal effect.

Theorem 4 (Calibration shatters the illusion)**.** *We observe an experimental study, J observational studies, and K calibration studies. Consider the model of Eqs. (3.24) to (3.29) and fit $\hat{\mu}, \hat{\gamma}^2$ from Eq. (3.30).*

The posterior $\mathrm{p}(\theta \,|\, y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \,;\, \hat{\mu}, \hat{\gamma}^2)$ is a normal distribution,

$$\theta \,|\, y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}, \mathbf{y}_{\mathrm{c}} \sim \mathcal{N}(\hat{\theta}_{\mathrm{CEB}}, \hat{v}^2_{\mathrm{CEB}}). \tag{3.31}$$

The posterior mean is

$$\hat{\theta}_{\mathrm{CEB}} = \frac{\sigma_{\mathrm{e}}^{-2} y_{\mathrm{e}} + \sum_{j=1}^{J} (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1} (y_{\mathrm{o},j} - \hat{\mu})}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1}}. \tag{3.32}$$

The posterior variance is

$$\hat{v}^2_{\mathrm{CEB}} = \left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} \left(\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2 \right)^{-1} \right)^{-1}. \tag{3.33}$$

Under the assumption that there exists a true prior $\mu^\star, \gamma^\star$, the estimator $\hat{\theta}_{\mathrm{CEB}}$ achieves asymptotically vanishing risk as we see more observational and calibration studies,

$$R(\hat{\theta}_{\mathrm{CEB}}, \theta^\star) \to 0 \quad \text{as } J, K \to \infty. \tag{3.34}$$

(See Theorem 6 in the Appendix for more results about the risk and consistency of the EB procedure.)

[1] As an extension, in Appendix B.2 we consider "internal" calibration studies that are paired with specific observational studies.

This theorem shows that we can use calibration studies to improve our estimator. In Eq. (3.30), we use maximum marginal likelihood, but there are other empirical Bayes methods. In our theory and simulations in Section 4, we also use moment matching. In particular, our choice of moment-matching estimators for γ^2 and μ are

$$\hat{\gamma}^2_{\mathrm{MM}} = \frac{1}{K}\sum_{k=1}^{K}\left\{(y_{\mathrm{c},k} - \bar{y}_{\mathrm{c}})^2 - \sigma^2_{\mathrm{c},k}\right\}, \qquad \bar{y}_{\mathrm{c}} = \frac{1}{K}\sum_{k=1}^{K} y_{\mathrm{c},k}, \tag{3.35}$$

and

$$\hat{\mu}_{\mathrm{MM}} = \frac{\sum_{k=1}^{K}(\hat{\gamma}^2_{\mathrm{MM}} + \sigma^2_{\mathrm{c},k})^{-1} y_{\mathrm{c},k}}{\sum_{k=1}^{K}(\hat{\gamma}^2_{\mathrm{MM}} + \sigma^2_{\mathrm{c},k})^{-1}}. \tag{3.36}$$

These moment-matching estimators enjoy the same asymptotic risk guarantees as in Theorem 2 and Theorem 4; see Proposition 3 and Proposition 4 in the Supplementary Material for details.

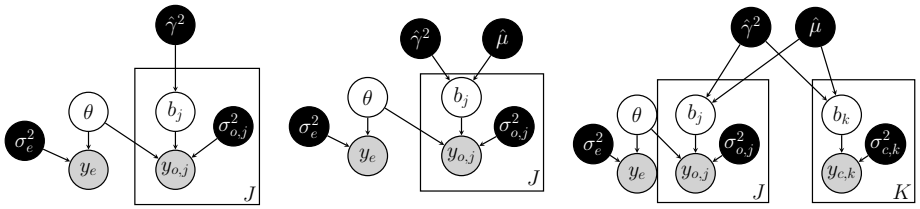


FIG 3.1. *Graphical representations of the hierarchical Bayesian models. The hats on the hyperparameters indicate that they are fitted by maximum marginal likelihood.* ***Left:*** *The empirical Bayes model in which the prior on the biases has mean zero and unknown variance. The illusion is shattered (Theorem 2).* ***Middle:*** *The empirical Bayes model in which the prior on the biases has unknown mean and variance. The illusion returns (Theorem 3).* ***Right:*** *The empirical Bayes model with calibration studies, in which the prior on the biases has unknown mean and variance. The illusion is shattered (Theorem 4).*

4. Simulations

Simulations help illustrate the theoretical findings in Section 3. In particular, we show that incorporating calibration studies leads to improved statistical efficiency and accuracy of causal effect estimation when combining experimental and observational studies, consistent with Theorem 4.

Setup and estimators. In each simulation, we generate one experimental study y_{e}, J observational studies $y_{\mathrm{o},1:J}$, and J calibration studies $y_{\mathrm{c},1:J}$. By default, we set the numbers of observational and calibration studies equal, so that $J = K$, though the same asymptotic properties continue to hold when

$J \neq K$; see Theorem 4. The data are generated as follows:

$$\begin{aligned} y_{\mathrm{e}} &\sim \mathcal{N}(\theta^\star, \sigma_{\mathrm{e}}^2), \\ b_j &\sim \mathcal{N}(\mu^\star, \gamma^{2\star}), \quad y_{\mathrm{o},j} \sim \mathcal{N}(\theta^\star + b_j, \sigma_{\mathrm{o}}^2), \\ b_{\mathrm{c},k} &\sim \mathcal{N}(\mu^\star, \gamma^{2\star}), \quad y_{\mathrm{c},k} \sim \mathcal{N}(b_{\mathrm{c},k}, \sigma_{\mathrm{c}}^2). \end{aligned}$$

Unless otherwise stated, the default parameters are $\theta^\star = 1.0$, $\mu^\star = 0.5$, $\gamma^{2\star} = 1.0$, and $\sigma_{\mathrm{e}} = \sigma_{\mathrm{o}} = \sigma_{\mathrm{c}} = 1.0$. We vary $J \in \{5, 10, 50, 100, 200, 500, 1000\}$ and use 10,000 simulation replicates per setting.

We compare the following estimators for $\theta^\star$:

- **Naive:** $\hat{\theta}_{\mathrm{e}} = y_{\mathrm{e}}$.
- **Calibrated EB:** Empirical Bayes moment matching estimator based on $(y_{\mathrm{e}}, y_{\mathrm{o},1:J}, y_{\mathrm{c},1:J})$.
- **Oracle:** Uses the true hyperparameters $(\mu^\star, \gamma^{2\star})$.

Evaluation metrics. We report the mean squared error

$$\mathrm{MSE}(J) := \mathbb{E}\left[(\hat{\theta} - \theta^\star)^2\right],$$

where $\hat{\theta}$ is estimated from J samples, and the relative efficiency (RE), defined as the ratio between the MSE of the calibrated empirical Bayes estimator and that of the experimental estimator,

$$\mathrm{RE}(J) = \frac{\mathrm{MSE}_{\mathrm{CEB}}(J)}{\mathrm{MSE}_{\mathrm{e}}(J)}.$$

Results. Figure 4.1 summarizes the results. The MSE of $\hat{\theta}_{\mathrm{e}}$ remains at σ_{e}^2 (grey dashed line).

The top-left panel shows that the calibrated EB estimator $\hat{\theta}_{\mathrm{CEB}}$ achieves asymptotically vanishing risk as J increases and converges to similar values across different noise levels σ_{e}^2. The linear association indicates a power-law relationship between the MSE and the sample size J, with slopes close to -1, indicating that the MSE scales as $O(J^{-1})$.

The top-right panel shows that the calibrated EB estimator $\hat{\theta}_{\mathrm{CEB}}$ scales as $O(J^{-1})$ for various levels of σ_{o}^2, with larger σ_{o}^2 inducing a higher MSE for a given sample size J.

The bottom-left panel shows that the MSE of $\hat{\theta}_{\mathrm{CEB}}$ scales as $O(J^{-1})$ for various specifications of θ. The MSE remains essentially unchanged across different values of θ for a fixed sample size J.

Finally, the bottom-right panel shows that the relative efficiency improves steadily with J, as the relative MSE decreases to 0, demonstrating the inferential gains from incorporating calibration studies.

While our calibrated EB estimates are based on the method of moments (3.35)–(3.36), the results based on maximum marginal likelihood are identical. We relay the results to Figure F.1 in the Appendix.

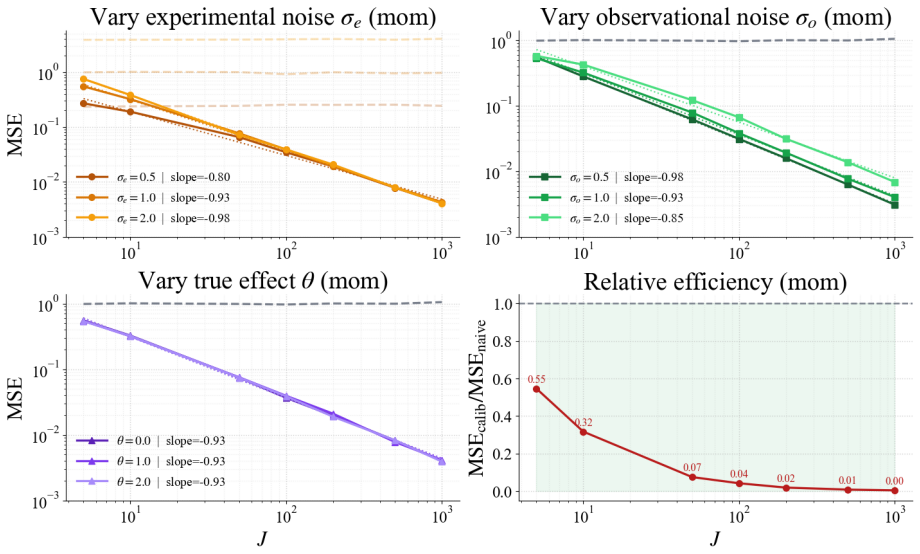


FIG 4.1. ***Top left:*** *Mean squared error (MSE) of the calibrated empirical Bayes estimator* $\hat{\theta}_{\mathrm{CEB}}$ *versus sample size* J*, for experimental noise levels* $\sigma_{\mathrm{e}} \in \{0.5, 1.0, 2.0\}$ *and fixed observational noise* $\sigma_{\mathrm{o}} = 1$*. The MSE decreases as a function of* J *and converges to the same value for large* J *for all values of* σ_{e}^2*.* ***Top right:*** *Comparison of MSE between the empirical Bayes estimator and the naive estimator as a function of observational noise* σ_{o}*, with experimental noise fixed at* $\sigma_{\mathrm{e}} = 1$*. The MSE decreases as a function of* J*. A smaller noise level* σ_{o}^2 *in the observational studies allows the MSE to decrease more sharply as the sample size* J *increases.* ***Bottom left:*** *MSE of both estimators across varying effect sizes* $\theta^\star$*, illustrating robustness to signal strength. The MSE scaling as a function of* J *is the same for all values of* $\theta^\star$*.* ***Bottom right:*** *Relative efficiency of the empirical Bayes estimator improves monotonically with* J*, dropping below* 0.5 *after* $J = 500$*. The relative efficiency of the calibrated empirical Bayes estimator converges to* 0 *as* J *increases.*

5. Real Data Example

In a large-scale field experiment conducted in 2007 by Ferraro and Miranda (2013) and colleagues in partnership with a water utility serving metropolitan Atlanta, Georgia, single-family households were randomly assigned either to a control group (approximately 71,800 households) or to one of three treatment arms (approximately 11,700 households in each). The three treatments were: (T1) a "tip sheet" mailed to households with information and suggestions on how to reduce outdoor and indoor water use (an information-only message); (T2) the same tip sheet plus a personalized letter that appealed to pro-social motivations (a weak social-norm message); (T3) the tip sheet plus the personalized letter and a social-comparison component, showing each household how its past water use compared with the median county usage (a strong social-norm message). The goal is to evaluate the effect of non-price messaging on residential water consumption during the summer watering season.

The results in Ferraro and Miranda (2013) show that T3 has the strongest effect out of the three arms, T2 has a modest effect, while T1 has a negligible effect.

We focus on a semisynthetic setting based on the T3 vs control group. Let $A_i = 1$ indicate that household i received the T3 message and $A_i = 0$ indicate not receiving any message. The outcome of interest O_i is the household's water usage during the 2008 summer season. We write $O_i(a)$ for the potential outcome under treatment and non-treatment $a \in \{0, 1\}$. The covariates X_i collect household information that could act as potential confounders including pre-treatment water use, demographics of the neighborhood and age of the house.

Our target estimand is the average treatment effect (ATE) of the T3 message on 2008 summer water usage,

$$\theta \;:=\; \mathbb{E}(O(1) - O(0)) \tag{5.1}$$

Because treatment assignment was randomized at the household level, the ignorability assumption (Assumption 1 in Appendix A.3 in the Supplementary Material) holds. The overlap assumption (Assumption 2 in Appendix A.3 in the Supplementary Material) is also satisfied since all households have positive probability to receive the treatment or control.

We specify the outcome model $\mu_a(X; \beta_a)$ as a linear regression model:

$$O_i \;=\; \alpha + \beta A_i + \delta X_i + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2), \tag{5.2}$$

where X_i may be a scalar or a vector. We assume, per Assumption 3, that this outcome model is correctly specified. Under Assumptions 1 to 3, the coefficient β identifies the target estimand θ (see Chapter 3.2 of Angrist and Pischke, 2009).

5.1. Constructing Experimental, Observational, and Calibration Estimators

We now design three semisynthetic datasets from the T3 data that correspond to the experimental, observational and calibration designs. First, we partition the

full sample into 100 disjoint subsets $B_1, \ldots, B_{100}$ via stratified sampling on the treatment and control groups to obtain approximately independent replicates. We designate the first subset B_1 as the *experimental* dataset, preserving the original randomized treatment assignment. For each remaining subset $B_2, \ldots, B_{100}$, we further construct two datasets: an *observational* dataset $B_{\mathrm{o},j}$, obtained by inducing covariate-dependent treatment assignment, and a *calibration* dataset $B_{\mathrm{n},j}$, obtained by introducing a pseudo-treatment that shares the same dependence on covariates but has no causal effect on the outcome.

Experimental Design. The experimental data inherits the randomized treatment assignment in the full data. For this reason, Assumptions 1 and 2 are satisfied; thus the ATE is fully identified. The ATE is exactly the experimental estimate at the population level:

$$\theta_e \;:=\; \mathbb{E}(O(1) - O(0)) \;=\; \theta \tag{5.3}$$

We estimate θ_e from the samples $B_{e,j}$ using the difference in means estimator, described in Example 1. Equivalently, we can use the OLS estimate $\hat{\beta}$ from the regression model (5.2), which provides an unbiased estimate of θ thanks to randomization.

Observational design. To simulate an observational study, we design $B_{\mathrm{o},j}$ so that treatment assignment is no longer randomized and instead depends on pre-treatment covariates. Specifically, we induce confounding by making the treatment assignment A_i a function of X_i, while keeping the outcome O_i generated according to the same structural outcome model as in the experimental design.

For the subset B_j, we modify its observational treatment assignment mechanism by specifying a new propensity score model $e_{\mathrm{o}}(X) := \mathbb{P}_{\mathrm{o}}(A = 1 \mid X)$ so that treatment assignment depends on the covariates X. For example, households with higher historical water consumption in summer 2006 are more likely to receive the T3 message. We reweight the samples in B_j so that the empirical distribution of $\{X_i, A_i\}_{i=1}^n$ matches $\{X_i, e_{\mathrm{o}}(X_i)\}_{i=1}^n$. This reweighting induces systematic differences in the distribution of X between treated and control units, which causes confounding by X on the causal effect of A.

Then, we intentionally *ignore* the confounding caused by X by using the difference in mean approach

$$\theta_{\mathrm{o}} \;:=\; \mathbb{E}_{\mathrm{obs}}(O \mid A = 1) - \mathbb{E}_{\mathrm{obs}}(O \mid A = 0).$$

Under the linear model (5.2) and the observational treatment mechanism,[2] we have

$$\theta_o = \beta + \delta\left(\mathbb{E}_{\mathrm{obs}}(X \mid A = 1) - \mathbb{E}_{\mathrm{obs}}(X \mid A = 0)\right) \tag{5.4}$$

$$= \theta + b \tag{5.5}$$

[2]Note that if the treatment were randomized, the true bias is zero but remains non-zero under $\mathbb{E}_{\mathrm{obs}}$.

where

$$b = \delta \left(\mathbb{E}_{\text{obs}}(X \mid A = 1) - \mathbb{E}_{\text{obs}}(X \mid A = 0) \right)$$

is the confounding bias induced by the imbalance in X between the treatment and control groups. Thus, the observational estimate θ_{o} is a bias estimate of the true causal effect θ.

Calibration design. Finally, we construct a calibration design $B_{\text{n},j}$ to quantify the bias term b. The key idea is to introduce a *pseudo-treatment* variable $\tilde{A}_i$ that has *no causal effect* on the outcome, so that the true ATE of $\tilde{A}$ on O is zero, but shares the *same dependence on* X as the observational treatment mechanism $e_{\text{o}}(\cdot)$.

Operationally, for each household $i \in B_j$ we generate

$$\tilde{A}_i \sim \text{Bernoulli}\big(e_{\text{o}}(X_i)\big), \tag{5.6}$$

independently of (A_i, O_i) given X_i.

The pseudo-treatment $\tilde{A}_i$ has no causal effect on O_i, as it is generated as a function of X_i. The ATE of $\tilde{A}$ on O therefore satisfies $\theta_{\text{n}}^{\text{true}} = 0$ by construction. However, $\tilde{A}_i$ is statistically associated with O_i through their shared dependence on X_i: households with covariate profiles that predict lower or higher water use are also more or less likely to have $\tilde{A}_i = 1$.

We define the calibration estimand as the difference in outcomes between the pseudo-treated and pseudo-control groups,

$$\theta_{\text{n}} := \mathbb{E}(O \mid \tilde{A} = 1) - \mathbb{E}(O \mid \tilde{A} = 0).$$

Under the model (5.2), the design of $e_{\text{o}}(\cdot)$, and the randomization in the original experiment, one can show that

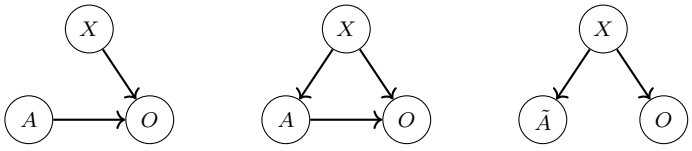
$$\theta_{\text{n}} \;=\; b, \tag{5.7}$$

so that the calibration estimand recovers precisely the bias term appearing in θ_{o}.

Intuitively, the contrast based on $\tilde{A}$ captures only the spurious association induced by covariate imbalance, rather than any genuine causal effect of T3, since $\tilde{A}$ has no direct effect on O. To estimate θ_{n}, we apply the difference in mean estimator for O_i on $\tilde{A}_i$ using $B_{\text{n},j}$.

Figure 5.1 summarizes the experimental, observational and calibration designs using causal graphs. In summary, for each subsample B_j we create

- $B_{e,j}$ to obtain an unbiased experimental estimate of the ATE θ;
- $B_{\text{o},j}$ to obtain a biased observational estimate $\theta_{\text{o}} = \theta + b$ that ignores the covariate-induced confounding; and
- $B_{\text{n},j}$ to obtain a calibration estimate $\theta_{\text{n}} = b$ that quantifies the bias.



(i) experiment (ii) observation (iii) calibration

FIG 5.1. ***Left:*** *the experimental design. The treatment assignment A is randomized and independent of X.* ***Middle:*** *the observational design. The arrow from X to A confounds the causal effect of A on O.* ***Right:*** *the calibration design. The causal relation from X to $\tilde{A}$ and from X to O are the same as (ii), but $\tilde{A}$ has no causal effect on O.*

5.2. Illusion vs Calibrated EB Models

The Illusion model from Section 3.4 assumes that observational studies, regardless of number or sample size, provide no additional information about θ due to unmeasured confounding. As a result, the posterior for θ does not contract with increasing data. By contrast, the calibrated empirical Bayes (CEB) model from Section 3.5 makes use of a negative control to facilitate bias estimation and correction of the observational estimates. The posterior under the calibrated EB model shows reduced uncertainty and more accurate estimates compared to the posterior under the Illusion model. The calibrated EB posterior is also more tightly distributed around the true parameter than the posterior based on a small subset of the experimental data, which illustrates how negative controls enable large observational datasets to be as or more informative than small RCTs.

Table 1 compares estimates from the Illusion and CEB models with those obtained from the reduced experimental subset B_1. The oracle benchmark is the ATE estimate and its 95% confidence interval constructed from the full experimental sample. The full-sample ATE estimate is -0.48, with a 95% confidence interval of $[-0.86, -0.10]$. In the subsequent error analysis, we treat the full-sample ATE as a proxy for the truth, that is, $\theta^\star \approx -0.48$.

The reduced experimental subset B_1 provides an ATE estimate of -1.40 with a wide confidence interval of $[-4.89, 2.08]$, which provides an imprecise reading of the oracle ATE that mimics the situation in Section 3.1. The Illusion posterior remains centered near the reduced experimental estimate but is substantially more concentrated, with a 95% credible interval of $[-1.84, -0.97]$. This pattern illustrates Theorem 3: observational data may appear to increase precision, resulting in overconfidence. In contrast, the calibrated EB posterior centers at -0.58, close to the oracle estimate of -0.48, and yields a tighter 95% credible interval of $[-0.90, -0.27]$. We fit the calibrated EB prior using the method-of-moments estimators in Eqs. (3.35) and (3.36)

We further assess uncertainty quantification by evaluating the coverage probability of the oracle 95% confidence interval under each posterior or sampling distribution. The Illusion model almost entirely misses the oracle interval, with

TABLE 1

Comparison of treatment effect estimates across models and experimental subsets. Values in brackets denote 95% credible or confidence intervals. The full experiment serves as an oracle benchmark for both point estimation and coverage assessment. The calibrated EB posterior achieves a point estimate close to the oracle and yields near-nominal coverage of the oracle 95% interval. In contrast, the Illusion posterior is overly concentrated around the reduced experimental estimate and exhibits severe under-coverage despite its apparently increased precision.

Method	Mean / 95% interval	$\Pr(\theta \in \mathrm{CI}_{\mathrm{full}})$	Rel. Prec. (vs Exp1)
Posterior (Calibrated EB)	−0.58 [−0.90, −0.27]	0.96	109.57
Posterior (Illusion)	−1.40 [−1.84, −0.97]	0.0068	4.44
Experiment (full)	−0.48 [−0.86, −0.10]	0.95	107.08
Experiment 1 (reduced)	−1.40 [−4.89, 2.08]	0.15	1.00

Notes. Relative precision is defined as the ratio of posterior expected mean squared errors relative to the reduced experiment 1, $\mathrm{Error}_{\mathrm{Exp1}}/\mathrm{Error}$. Values > 1 indicate higher precision than the single reduced experiment. "Experiment (full)" uses the full sample; missing values indicate quantities not applicable. The column $\Pr(\theta \in \mathrm{CI}_{\mathrm{full}})$ reports the posterior or sampling coverage of the 95% confidence interval of $\theta^\star$ constructed from the full experimental sample.

coverage probability 0.0068, while the reduced experiment captures it with probability 0.15. In contrast, the calibrated EB model achieves well-calibrated uncertainty, with coverage probability 0.96, slightly above the nominal level. Finally, we compare relative precision (inverse risk) against the reduced experiment. The calibrated EB model attains a relative precision of 109.57, indicating much lower error than the reduced experiment, whereas the Illusion model attains a relative precision of 4.44. The apparent gain under the Illusion model comes at the cost of severely miscalibrated uncertainty.

Figure 5.2 visualizes these differences. Under the reduced experiment with a flat prior (grey), the posterior is diffuse, reflecting substantial uncertainty, and its mean (green dashed line) deviates from the full-sample estimate (red). Under the Illusion model (green), the posterior concentrates sharply around the reduced experimental estimate but largely fails to overlap the full-sample 95% confidence interval (red shaded region). In contrast, under the CEB model in Eqs. (3.24) to (3.29), the posterior mean (purple dashed line) closely tracks the full-sample estimate and the posterior mass overlaps the oracle confidence interval, which suggests both accurate point estimation and calibrated uncertainty.

6. Discussion

We study how to use observational studies to improve a causal estimate from an experiment. We take an empirical Bayes perspective on this problem. When only an experimental study and a collection of observational studies are available, and we make no assumptions about the distribution of bias, we encounter the "illusion of learning" (Gerber, Green and Kaplan, 2004): the observational studies may look relevant, but they do not change the posterior estimate of the causal effect.

One way to break the illusion is to assume that the biases of observational studies have mean zero in the population. Under this assumption, the observa-

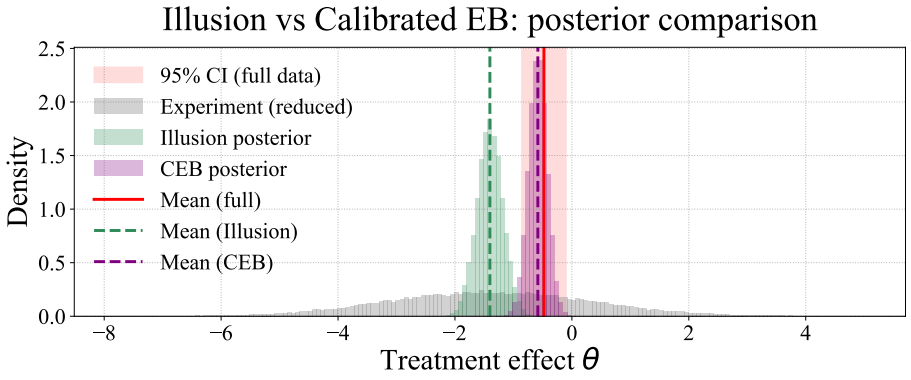


FIG 5.2. *Posterior distributions of the causal effect θ under three models. The reduced experimental posterior (grey) is diffuse, reflecting limited information. The Illusion posterior (green) is overly concentrated around the reduced experimental estimate and fails to overlap with the full 95% confidence interval (red shaded region). The Calibrated EB posterior (purple) is centered near the full experimental ATE (red line) and shows well-calibrated uncertainty.*

tional studies meaningfully affect the estimate, and the risk goes to zero as the number of studies grows. But this assumption is often implausible. We therefore introduce calibration studies to learn the full distribution of bias. Our theory shows that, with enough observational and calibration studies, the empirical Bayes estimates of the bias distribution are consistent, and so is the resulting posterior estimate of the causal effect. The simulations and semi-synthetic analysis illustrate the idea.

Although strong, versions of the zero-mean assumption appear, directly or indirectly, in parts of the research literature. In the social sciences, for example, arguments for multi-method research and triangulation often rest on the idea that different methods have different weaknesses, and that these weaknesses may offset one another. Brewer and Hunter (2006) write that the flaws of different methods are "not identical," so combining them can help compensate for their limitations. Brady and Collier (2010) make a similar point in arguing that qualitative and quantitative approaches can address one another's shortcomings. Summarizing triangulation, Creswell and Creswell (2018) observes that researchers have often hoped that biases in one method could cancel biases in another. Our analysis clarifies what these conjectures require: to gain from combination, one must assume or learn something about how biases are distributed across observational studies.

One implication of our final model is that calibration studies are especially valuable. Not only do they help quantify the bias of an observational research approach; in doing so, they enable an observational study using that approach to contribute meaningfully to the estimation of the target causal parameter. In a domain where observational studies abound, calibration studies have the potential to unlock otherwise uninterpretable research findings. That said, we

emphasize that this strategy depends on calibration studies being subject to the same population of biases as the target observational studies.

Calibration studies can take different forms in practice. Consider first a social science example about the effect of phone calls that encourage people to vote. An observational study might estimate this effect with regression or matching, comparing turnout among people who receive calls to turnout among untreated people with similar background covariates. This design is vulnerable to selection bias if, even after conditioning on covariates, potential outcomes differ systematically between those who receive calls and those who do not. For example, people with a higher propensity to vote may also be more likely to answer the phone. A calibration study can assess this bias by randomly assigning placebo calls that have nothing to do with the upcoming election versus no call at all. Applying the same regression or matching analysis to this placebo intervention reveals how much bias the observational design introduces; see, for example, Arceneaux, Gerber and Green (2010) and Gerber et al. (2010).

This biomedical example also uses a placebo design in order to gauge the effects of hormone therapy on a variety of medical outcomes, including mortality (Curtis et al., 2011). The observational analysis gauges the effects of therapy by comparing high-adherence subjects (those who took the prescribed dosage) with low-adherence subjects. Adherence to the hormone treatment seemed to reduce risk of mortality, even after adjusting for covariates. However, the same relationship also held among those who adhered to the placebo arm; the apparent "effect" of adherence among those in the placebo arm identifies the bias inherent in the comparison of high-adherence and low-adherence subjects. In both of these examples, the bias of an estimation strategy is revealed when it is applied to a placebo intervention whose effect is assumed to be zero.

Methodologically, there are several interesting directions to extend this work. One is to extend our EB approach to nonparametric priors. The normal prior assumption $b_j \overset{iid}{\sim} \mathcal{N}(\mu, \gamma^2)$ can be relaxed. Suppose instead that $b_j \overset{iid}{\sim} g(b)$ for an unknown prior distribution g. We can fit g from calibration studies using nonparametric maximum likelihood (Kiefer and Wolfowitz, 1956; Jiang and Zhang, 2009; Efron, 2014; Soloff, Guntuboyina and Sen, 2025). Another direction is to move beyond the normal means model. The same empirical Bayes idea should extend to regression settings and other hierarchical models (Gelman and Hill, 2006). Finally, a third avenue of future work is to characterize the tradeoff between conducting experimental, observational, and calibration studies when estimating a target causal effect.

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Appendix A: Within-Study Causal Estimation

In our approach, we have treated y_e, y_{o}, and y_{n} as observations, but in fact they are estimators of the causal target θ computed from experimental and observational datasets. Here, we clarify the assumptions and methods used to obtain these estimates.

Each study provides a dataset from which we estimate causal effects using the potential outcomes framework. Suppose the treatment is a binary variable $A \in \{0, 1\}$, where $A = 0$ denotes control and $A = 1$ denotes treatment. We assume the existence of potential outcomes $O(0)$ and $O(1)$, representing the outcomes that would be observed under control and treatment, respectively. The observed outcome is given by $O = AO(1) + (1 - A)O(0)$. Let (X, Z) denote a vector of pre-treatment covariates and instruments, where X is observed but potentially confounded with O, and Z affects O only through X. The available dataset is $\mathcal{D}_n = \{A_i, X_i, Z_i, O_i\}_{i=1}^n$. We also posit unobserved confounders U_i, which are not available for any unit. We assume a superpopulation model in which the full-data variables $\mathcal{D} := \{A_i, X_i, Z_i, O_i(1), O_i(0)\}_{i=1}^n$ are i.i.d. draws from a population distribution.

The goal is to estimate either the average treatment effect (ATE) $\theta = \mathbb{E}[O(1) - O(0)]$ or the conditional ATE $\theta = \mathbb{E}[O(1) - O(0) \mid X]$. Because of the i.i.d. assumption, we drop unit indices in the following expressions. Define the outcome regression functions as $\mu_a(X) = \mathbb{E}[O \mid A = a, X]$ and the propensity score as $e(X) = \mathbb{P}(A = 1 \mid X)$.

A.1. Identification

The fundamental problem of causal inference is that at most one of $O(1)$ or $O(0)$ is available for a unit. Following Rosenbaum and Rubin (1983), we lay out the identification assumptions to identify causal effects.

Assumption 1 (Ignorability)**.** $O(1), O(0) \perp\!\!\!\perp A \mid X$.

Under ignorability, treatment assignment is conditionally independent of potential outcomes given X. It states that conditional on the observed characteristics, selection bias disappears.

Moreover, we require sufficient overlap between the treatment and control covariate distributions, stated in terms of the *propensity score* $e(X)$.

Assumption 2 (Overlap)**.** *There exist constants* c_1, c_2 *such that with probability one,* $0 < c_1 \leq e(X) \leq c_2 < 1$.

Under ignorability and overlap, the ATE θ can be identified and estimated through regression imputation, inverse probability weighting (IPW), augmented inverse probability weighting (AIPW), or matching (See the descriptions in Appendix A.3).

In practice, the outcome distribution and the propensity score are typically unknown and must be estimated based on a model. We require them to be well-specified.

Assumption 3 (Outcome model)**.** *The parametric model $\mu_a(X;\beta_a)$ correctly specifies $\mu_a(X)$ for $a \in \{0,1\}$; that is, $\mu_a(X) = \mu_a(X;\beta_a^*)$, where β_a^* denotes the true parameter.*

To ensure the outcome model is well specified, we use regression with a rich set of covariates, including all possible interactions among them. This strategy effectively blocks backdoor paths through X.

Assumption 4 (Propensity score model)**.** *The parametric model $e(X;\alpha)$ correctly specifies $e(X)$; that is, $e(X) = e(X;\alpha^*)$, where α^* denotes the true model parameter.*

Observational data. The key difference between observational and experimental data is that observational data are generally confounded; conditioning on X alone does not render treatment independent of outcomes. Instead, there exist unobserved confounders $U \in \mathcal{U}$ that jointly influence both A and O.

Assumption 5 (Ignorability and overlap for observational data)**.** *For $a \in \{0,1\}$, $O(a) \perp\!\!\!\perp A \mid X, U$, and $0 < e(X,U) < 1$ almost surely.*

This assumption implies that U and X together account for all confounding in the observational data, and generally observing X alone is insufficient. Consequently, the ATE is not identifiable with the observed X alone.

A.2. Estimating the Variances

In our model, we assume that the variances $\sigma^2_{\mathrm{o},1:J}$ and $\sigma^2_{\mathrm{n},1:K}$ are fixed and known. In practice, these variances must be estimated from the data in each study.

When the estimands corresponding to y_e and $y_{\mathrm{o},j}$ admit a functional form as $\mathcal{F}(\mathcal{D}_n)$, the variances of these estimators-such as σ_e^2 can often be derived from their exact limiting distributions (Imbens and Rubin, 2015) the asymptotic semiparametric efficiency bound (van der Laan and Rose, 2011). Alternatively, one can estimate variances using a parametric bootstrap procedure: resample the data within each study with replacement and compute the sample variance of the estimate $\hat{\theta}$ over the resampled datasets $\mathcal{D}_n^{(1)}, \ldots, \mathcal{D}_n^{(B)}$.

A.3. Review of Important Estimators

We briefly review several commonly used causal estimators. In our setting, for each unit j, the potential outcome under the observed treatment A_j is O_j, while the (counterfactual) potential outcome under $1 - A_j$ is unobserved. This missing potential outcome can be imputed, for example, using the average of observed outcomes among the M nearest units receiving the opposite treatment $(1 - A_j)$.

Example 1 (Difference in means)**.** *Let $n_a = \sum_{j=1}^{n} \mathbb{I}(A_j = a)$ denote the number of units receiving treatment $a \in \{0,1\}$. A simple estimator of the ATE is the*

difference-in-means estimator,

$$\hat{\theta}_{DIFF} = \frac{1}{n_1} \sum_{j:A_j=1} O_j - \frac{1}{n_0} \sum_{j:A_j=0} O_j.$$

Under a completely randomized experiment, the difference-in-means estimator is unbiased for the ATE. It is asymptotically normal under mild conditions (Ding, 2024, Theorem 4.2), and its variance can be estimated either by the bootstrap or by an explicit yet slightly conservative variance upper bound (Neyman, 1923).

Example 2 (Matching estimators)**.** *To fix ideas, consider matching with replacement, with the number of matches fixed at M. The method of matching imputes the missing potential outcomes for each unit. Let the matched units for unit j be indexed by S_j, determined, for instance, by Euclidean distance. Define the imputed potential outcome as $\hat{O}_j(a) = O_j \mathbb{I}(a = A_j) + \frac{1}{M} \sum_{k \in S_j} O_k (1 - \mathbb{I}(a = A_j))$. The matching estimator of the ATE θ is then given by*

$$\hat{\theta}_{MAT} = \frac{1}{n_1} \sum_{j:A_j=1} \left(O_j - \frac{1}{M} \sum_{k \in S_j} O_k \right).$$

Since nearest-neighbor matching introduces a non-vanishing bias, two strategies can be employed to address this issue: one is to debias the matching estimator using the approach of Abadie and Imbens (2012); alternatively, the bootstrap method can be used to estimate its variance when asymptotic approximations are unreliable.

Example 3 (Inverse-probability-weighted estimator for the ATE)**.** *Dehejia and Wahba (1999) use propensity-score matching to estimate causal effects. For the ATE, the inverse-probability-weighted (IPW) estimator is given by*

$$\hat{\theta}_{PSM} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{A_i \, O_i}{e(X_i; \hat{\alpha})} - \frac{(1 - A_i) \, O_i}{1 - e(X_i; \hat{\alpha})} \right),$$

where A_i is the treatment indicator, O_i is the observed outcome, and $e(X_i; \hat{\alpha})$ denotes the estimated propensity score model. This estimator reweights observed outcomes by the inverse of their estimated treatment probabilities, thereby balancing the covariate distributions between treated and control groups. The IPW estimator is shown to be asymptotically normal (Hirano, Imbens and Ridder, 2003, Theorem 1) and its variance can be estimated by the bootstrap.

Appendix B: Extensions

B.1. Resource allocation.

Previously, we studied the scenario where a researcher forms their prior beliefs after collecting results from experimental and observational studies. Now, we

turn to a related question: how should this researcher allocate resources between experimental and observational studies?

Suppose the research has a total budget of R. The marginal cost of an experimental study, observational study and calibration study are π_e, π_o, π_c. Let n_e, n_o, n_c be the numbers of the experimental, observational, and calibration studies. The allocation is subject to the budget constraint:

$$\pi_e n_e + \pi_o n_o + \pi_c n_c = R. \tag{B.1}$$

Suppose the variance of each experimental study is σ_e^2, and the variance for the observational study is σ_o^2. Using previous results, the goal is to allocate resources to minimize the posterior variance of θ, subject to the budget constraint R.

The goal is to minimize the empirical Bayes posterior variance $\hat{\sigma}^2_{EB}$, or equivalently maximizing $1/\hat{\sigma}^2_{EB}$, subject to the the budget constraint (B.1).

$$\max_{n_e, n_o, n_c} \left(n_e \sigma_e^{-2} + \sum_{j=1}^{n_o} \left(\sigma^2_{o,j} + \hat{\gamma}^2 \right)^{-1} \right) \quad \text{s.t. } \pi_e n_e + \pi_o n_o + \pi_c n_c = R. \tag{B.2}$$

The resource allocation problem (B.2) is NP-hard, as it reduces to an integer program with an exponentially large search space. When $\sigma^2_{o,j} = \sigma^2_o$ for all j and $n_c = 0$, problem (B.2) admits exact solutions provided by (Gerber, Green and Kaplan, 2004, Theorem 2).

A scalable approximate solution to problem (B.2) is given below.

(1) Since $\hat{\gamma}^2$ depends only on n_c, we can treat n_c as fixed, reduce the budget to $R' = R - \pi_c n_c$, and solve the subproblem

$$\max_{n_e,\, z_1, \ldots, z_J} \left\{ n_e \sigma_e^{-2} + \sum_{j=1}^{J} z_j \left(\sigma^2_{o,j} + \hat{\gamma}^2(n_c) \right)^{-1} \right\} \quad \text{s.t. } \pi_e n_e + \pi_o \sum_{j=1}^{J} z_j \le R'.$$

This is a knapsack problem with one additional integer variable n_e.

(2) Relax the integrality constraints to $n_e \ge 0, 0 \le z_j \le 1$. The relaxed problem is a linear program and has the explicit greedy solution:

- Compute the *value-per-cost ratios*

$$\rho_e := \frac{\sigma_e^{-2}}{\pi_e}, \qquad \rho_j := \frac{\left(\sigma^2_{o,j} + \hat{\gamma}^2(n_c) \right)^{-1}}{\pi_o} \quad (j = 1, \ldots, J).$$

- Sort observational units so that $\rho_{(1)} \ge \rho_{(2)} \ge \cdots \ge \rho_{(J)}$.
- In the relaxed problem, the optimal solution allocates the entire remaining budget R' to the option (experimental or observational) with the highest ρ and, if needed, partially fills the last item. Concretely, the relaxed optimizer is of the form

$$(n_e^{\text{rel}}, z_1^{\text{rel}}, \ldots, z_J^{\text{rel}}) = \begin{cases} \left(R'/\pi_e,\, 0, \ldots, 0 \right), & \text{if } \rho_e \ge \rho_{(1)}, \\ \left(0,\, 1, \ldots, 1,\, \tilde{z}_{(k)},\, 0, \ldots, 0 \right), & \text{if } \rho_{(k)} > \rho_e \text{ for some } k, \end{cases}$$

where $\tilde{z}_{(k)} \in (0, 1)$ is chosen to exhaust the budget.

(3) Transform the relaxed solution into a feasible integer solution by
   - Rounding n_e^{rel} down to $n_e = \lfloor n_e^{\text{rel}} \rfloor$;
   - Taking $z_{(1)} = \cdots = z_{(k-1)} = 1$, $z_{(k)} = \mathbf{I}\{\text{enough remaining budget}\}$, and $z_{(j)} = 0$ for $j > k$;
   - If budget remains, optionally increment n_e by 1 as long as π_e fits.

   This yields a feasible integer solution whose objective value is within a constant-factor of the optimal knapsack value (a standard guarantee for greedy rounding).

(4) Starting from the rounded solution, perform a simple local search: repeatedly consider "swap" moves that replace one experimental unit by an observational unit (or vice versa) while respecting the budget and increasing the objective. This local search terminates at a 1-step optimal solution and typically improves the greedy baseline.

(5) Repeat steps (1)–(4) over a discrete grid $n_c \in \{0, 1, \ldots, n_c^{\max}\}$ and select the triplet (n_e, n_o, n_c) with the largest value of (B.2).

This iterative procedure avoids exhaustive search over the exponentially large integer grid by exploiting the knapsack structure of the problem.

B.2. External and internal calibrations.

There are two types of calibration studies: *external* and *internal*. An external calibration study provides information about the population of biases, particularly their variance. This helps us improve the estimation of the causal effect with observational data by learning the prior of biases. An internal calibration study is paired with a specific observational study and is subject to the same bias but unrelated to the causal effect θ. Specifically, if we observe y_o from a treatment effect θ with bias b, then the corresponding internal calibration study provides an observation subject to the same b but independent of θ. These studies allow us to estimate b directly rather than infer it only through a prior, thereby enabling the use of observational data to estimate θ.

So far, our analysis has focused on *external* calibration studies $\boldsymbol{y}_{c,1}, \ldots, \boldsymbol{y}_{c,K}$ that are unpaired with observational studies. In contrast, *internal calibration* assumes access to pairs of calibration and observational studies. Each internal calibration study is designed to quantify the specific bias associated with a particular observational study. The goal is to estimate the ATE within each study rather than for a broader meta-analytic superpopulation:

$$y_{o,j} \sim \mathcal{N}(\theta + b_j, \sigma_{o,j}^2), \quad y_{c,j} \sim \mathcal{N}(b_j, \sigma_{c,j}^2), \quad \text{for } j \in [J].$$

Figure B.1 represents the generative process for this model.

Although each calibration study $y_{c,j}$ is directly paired with an observational study, we adopt an empirical Bayes approach to estimating the biases.

To calibrate biases using internal calibration, we seek an estimator $\hat{\boldsymbol{b}} := (\hat{b}_1, \ldots, \hat{b}_J)$ of the bias vector $\boldsymbol{b} := (b_1, \ldots, b_J)$ from the calibration studies $y_{c,1:J}$

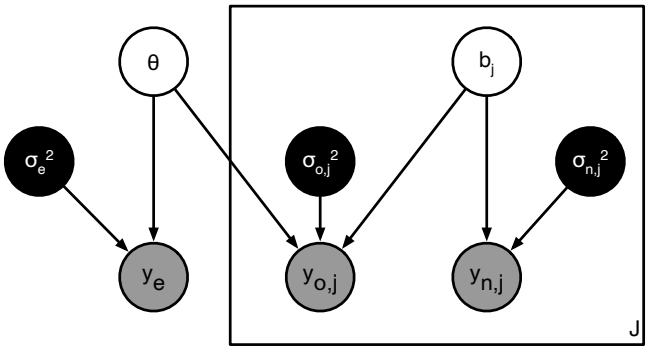


FIG B.1. *The hierarchical model with internal calibration studies.*

under the model $y_{c,j} \sim \mathcal{N}(b_j, \sigma^2_{c,j})$. This corresponds to the normal means problem with heteroskedastic noise.

While no explicit prior is shown in Figure B.1, the empirical Bayes formulation arises naturally when the objective is to minimize $\|\hat{\boldsymbol{b}} - \boldsymbol{b}\|_2^2$. Consider the classical conjugate model with $b_j \sim \mathcal{N}(\mu, \gamma^2)$ and unknown hyperparameters μ and γ^2. As shown by Xie, Kou and Brown (2012), the asymptotically optimal shrinkage estimator is

$$\hat{b}_j = \frac{\hat{\gamma}^2}{\sigma^2_{c,j} + \hat{\gamma}^2} y_{c,j} + \frac{\sigma^2_{c,j}}{\sigma^2_{c,j} + \hat{\gamma}^2} \hat{\mu}, \tag{B.3}$$

where $(\hat{\mu}, \hat{\gamma}^2)$ minimize the Stein unbiased risk estimate (SURE):

$$(\hat{\mu}, \hat{\gamma}^2) \in \underset{\gamma^2 \geq 0,\, \mu \in \mathbb{R}}{\arg\min} \frac{1}{J} \sum_{j=1}^{J} \frac{\sigma^2_{c,j}}{(\sigma^2_{c,j} + \gamma^2)^2} \left[\sigma^2_{c,j} (y_{c,j} - \mu)^2 + \gamma^2 - \sigma^2_{c,j} \right]. \tag{B.4}$$

To estimate the causal parameter θ with internal validation studies, we could use the following *empirical Bayes estimator*, which plugs in the shrinkage estimates $\hat{b}_1, \ldots, \hat{b}_J$:

$$\hat{\theta}_{\text{EB}} := \frac{\sigma_e^{-2} y_e + \sum_{j=1}^{J} \sigma_{o,j}^{-2} (y_{o,j} - \hat{b}_j)}{\sigma_e^{-2} + \sum_{j=1}^{J} \sigma_{o,j}^{-2}}. \tag{B.5}$$

Appendix C: Support Results

The posterior under the Gaussian bias model admits a closed-form expression.

Proposition 1. *Consider the model*

$$\theta \sim \text{p}(\theta) \propto 1, \tag{C.1}$$

$$y_e \sim \mathcal{N}(\theta, \sigma_e^2), \tag{C.2}$$

$$b_j \sim \mathcal{N}(\mu, \gamma^2), \qquad j \in [J], \tag{C.3}$$

$$y_{o,j} \sim \mathcal{N}(\theta + b_j, \sigma^2_{o,j}). \tag{C.4}$$

Then the posterior of θ is Gaussian with mean μ_{post} and variance σ^2_{post}, where

$$\mu_{post} = \frac{\sigma_{\mathrm{e}}^{-2} y_{\mathrm{e}} + \sum_{j=1}^{J} (\sigma^2_{\mathrm{o},j} + \gamma^2)^{-1} (y_{\mathrm{o},j} - \mu)}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma^2_{\mathrm{o},j} + \gamma^2)^{-1}}, \quad \sigma^2_{post} = \left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma^2_{\mathrm{o},j} + \gamma^2)^{-1} \right)^{-1}.$$

Proof. After marginalizing out the latent biases $b_1, \ldots, b_J$, the variables y_{e} and $y_{\mathrm{o},1:J}$ are conditionally independent given θ,

$$\begin{aligned} \theta &\sim \mathrm{p}(\theta) \propto 1, \\ y_{\mathrm{e}} &\sim \mathcal{N}(\theta, \sigma^2_{\mathrm{e}}), \\ y_{\mathrm{o},j} &\sim \mathcal{N}\left(\theta + \mu, \gamma^2 + \sigma^2_{\mathrm{o},j}\right), \qquad j \in [J]. \end{aligned}$$

The log-likelihood of the model above is

$$\log \mathrm{p}(y_{\mathrm{e}}, y_{\mathrm{o},1:J} \mid \theta) = -\frac{(y_{\mathrm{e}} - \theta)^2}{2\sigma^2_{\mathrm{e}}} - \sum_{j=1}^{J} \frac{(y_{\mathrm{o},j} - \mu - \theta)^2}{2(\sigma^2_{\mathrm{o},j} + \gamma^2)} + \text{const.}$$

Assuming a flat prior $\mathrm{p}(\theta) \propto 1$, the posterior is proportional to the likelihood, so

$$\log \mathrm{p}(\theta \mid y_{\mathrm{e}}, y_{\mathrm{o},1:J}) \overset{c}{=} -\frac{1}{2} \left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma^2_{\mathrm{o},j} + \gamma^2)^{-1} \right) \theta^2 + \left(\frac{y_{\mathrm{e}}}{\sigma^2_{\mathrm{e}}} + \sum_{j=1}^{J} \frac{y_{\mathrm{o},j} - \mu}{\sigma^2_{\mathrm{o},j} + \gamma^2} \right) \theta.$$

This identifies the posterior as a normal distribution with the stated mean μ_{post} and variance σ^2_{post}. □

Appendix D: Complete Theorems

In this section, we provide the full statements of the theorems from Sections 3.3 and 3.5. Let $\mathbb{P}_{\theta,\mu,\gamma^2}$ be the hierarchical model in Eq. (C.1) with parameters θ, μ, and γ^2.

In each setting, we assume there exist true prior parameters $\gamma^{2\star} \geq 0$ and $\mu^\star \in \mathbb{R}$ (if included) and a causal effect $\theta^\star \in \mathbb{R}$ that generate the studies, and that the true data-generating distribution is $\mathbb{P}^\star = \mathbb{P}_{\theta^\star,\mu^\star,\gamma^{2\star}}$.

For our theory, we first prove a general risk consistency theorem for any consistent data-driven estimator $\hat{\gamma}^2$ of the prior variance, using the technique of sample splitting. We then show that the estimator $\hat{\gamma}^2$ fit by maximum marginal likelihood and the estimator $\hat{\gamma}^2_{\mathrm{MM}}$ fit by moment matching satisfy the risk consistency result under additional regularity assumptions.

Empirical Bayes with zero-mean bias assumption.

Theorem 5. *Let $y_{\mathrm{o},1:2J}$ be the observational estimates, and suppose the zero-mean bias assumption holds, i.e., $\mu^\star = 0$. Let $\hat{\gamma}^2$ be an estimator of $\gamma^{2\star}$ constructed from $y_{\mathrm{o},(J+1):2J}$ such that*

$$\mathbb{E}_{\mathbb{P}^\star}\left[\left|\hat{\gamma}^2 - \gamma^{2\star}\right|\right] \to 0 \qquad \text{as } J \to \infty.$$

Assume moreover that

$$\sum_{j=1}^{J} \sigma_{\mathrm{o},j}^{-2} \to \infty \qquad \text{as } J \to \infty.$$

Then the empirical Bayes estimator $\hat{\theta}_{\mathrm{EB}_0}$ *in Eq.* (3.16) *achieves asymptotically vanishing risk:*

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) \to 0 \qquad \text{as } J \to \infty.$$

Crucially, Theorem 5 requires $\hat{\gamma}^2$ to be consistent in the L^1 sense. Below, we provide sufficient conditions under which the maximum likelihood estimator and the moment-matching estimator of γ^2 satisfy this requirement.

Proposition 2 (Consistency of the maximum likelihood estimator)**.** *Suppose that* $\gamma^{2\star} \in [0, B]$ *for some fixed* $B < \infty$*. Let*

$$\hat{\gamma}^2 \in \operatorname*{arg\,max}_{\gamma^2 \in [0,B]} \log \mathrm{p}(y_{\mathrm{e}}, y_{\mathrm{o},(J+1):2J}\,;\, \gamma^2).$$

Suppose

$$\sum_{j=J+1}^{2J} \sigma_{\mathrm{o},j}^{-2} \to \infty \qquad \text{and} \qquad \sup_{j \geq 1} \sigma_{\mathrm{o},j}^2 < \infty \qquad \text{as } J \to \infty.$$

Then $\hat{\gamma}^2 \to \gamma^{2\star}$ *in probability and* $\mathbb{E}_{\mathbb{P}_{\theta^\star,0,\gamma^{2\star}}}\left[|\hat{\gamma}^2 - \gamma^{2\star}|\right] \to 0$ *as* $J \to \infty$*.*

Proposition 3 (Consistency of the moment-matching estimator)**.** *Suppose that*

$$\frac{1}{J^2} \sum_{j=J+1}^{2J} \sigma_{\mathrm{o},j}^4 \to 0 \qquad \text{as } J \to \infty.$$

Let $\hat{\gamma}^2_{\mathrm{MM}}$ *be the moment-matching estimator defined as*

$$\hat{\gamma}^2_{\mathrm{MM}} = \frac{1}{J} \sum_{j=J+1}^{2J} \left(y_{\mathrm{o},j}^2 - \sigma_{\mathrm{o},j}^2\right).$$

Then $\hat{\gamma}^2 \to \gamma^{2\star}$ *in probability and* $\mathbb{E}_{\mathbb{P}_{\theta^\star,0,\gamma^{2\star}}}\left[|\hat{\gamma}^2 - \gamma^{2\star}|\right] \to 0$ *as* $J \to \infty$*.*

Empirical Bayes with calibration studies. Suppose we also have K calibration studies, $y_{\mathrm{c},1:K} = \{y_{\mathrm{c},1}, \ldots, y_{\mathrm{c},K}\}$, as in the setting of Section 3.5. Assume that the biases in the calibration studies follow the same distribution as the biases in the observational studies:

$$\begin{aligned} b_{\mathrm{c},k} &\overset{iid}{\sim} \mathcal{N}(\mu^\star, \gamma^{2\star}), \qquad k = 1, \ldots, K, \\ y_{\mathrm{c},k} &\sim \mathcal{N}(b_{\mathrm{c},k}, \sigma_{\mathrm{c},k}^2). \end{aligned} \tag{D.1}$$

Given γ^2, the likelihood is concave in μ, so the profiled MLE for μ is

$$\hat{\mu}(\gamma^2) = \frac{\sum_{k=1}^{K}(\gamma^2 + \sigma_{\mathrm{c},k}^2)^{-1} y_{\mathrm{c},k}}{\sum_{k=1}^{K}(\gamma^2 + \sigma_{\mathrm{c},k}^2)^{-1}}. \tag{D.2}$$

If we have an estimate $\hat{\gamma}^2$ of the prior variance, then a natural estimator of μ is the plug-in profiled MLE $\hat{\mu} = \hat{\mu}(\hat{\gamma}^2)$.

Theorem 6. *Consider the calibrated model with experimental, observational, and calibration studies. Suppose the following conditions hold as $J, K \to \infty$:*

1. $\sum_{j=1}^{J} \sigma_{\mathrm{o},j}^{-2} \to \infty$,
2. $\left(\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{-2}\right) \min_{k\in[K]} \sigma_{\mathrm{c},k}^{4} / \log K \to \infty$,
3. $\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{-4} \to \infty$,
4. $\dfrac{\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{-4}}{\sum_{k=1}^{K}(B + \sigma_{\mathrm{c},k}^2)^{-2}} = O(1)$ *for any* $B > 0$,
5. $\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{-2} / (\sqrt{K} \log K) \to \infty$,
6. $\dfrac{\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{-2}}{\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{2}} = O(1)$.

Let $\hat{\gamma}^2$ be a consistent estimator of $\gamma^{2\star}$ based on $y_{\mathrm{c},1:K}$ such that

$$\mathbb{E}_{\mathbb{P}^\star}\left[|\gamma^{2\star} - \hat{\gamma}^2|\right] \to 0 \qquad \text{as } K \to \infty,$$

and let $\hat{\mu}$ be the profiled MLE in Eq. (D.2) evaluated at $\hat{\gamma}^2$. Then the empirical Bayes posterior of θ is a Normal distribution with mean $\hat{\theta}_{\mathrm{CEB}}$ and variance $\hat{v}_{\mathrm{CEB}}^2$, where

$$\hat{\theta}_{\mathrm{CEB}} = y_{\mathrm{e}} + \frac{\sum_{j=1}^{J}(\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}(y_{\mathrm{o},j} - \hat{\mu} - y_{\mathrm{e}})}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J}(\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}, \quad \hat{v}_{\mathrm{CEB}}^2 = \left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J}(\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}\right)^{-1}.$$

Moreover, the EB estimator $\hat{\theta}_{\mathrm{CEB}}$ satisfies

$$R(\hat{\theta}_{\mathrm{CEB}}, \theta^\star) \to 0 \qquad \text{as } J, K \to \infty.$$

Below, we provide sufficient conditions for the the moment-matching estimator of γ^2 satisfy consistency in the L_1 sense.

Proposition 4 (Consistency of the moment-matching estimator)**.** *Suppose that*

$$\frac{1}{K^2}\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^4 \to 0 \qquad \text{as } K \to \infty.$$

Let $\hat{\gamma}_{\mathrm{MM}}^2$ be the moment-matching estimator defined by

$$\hat{\gamma}_{\mathrm{MM}}^2 = \frac{1}{K}\sum_{k=1}^{K}\left\{(y_{\mathrm{c},k} - \bar{y}_{\mathrm{c}})^2 - \sigma_{\mathrm{c},k}^2\right\}, \qquad \bar{y}_{\mathrm{c}} = \frac{1}{K}\sum_{k=1}^{K} y_{\mathrm{c},k}.$$

Then $\hat{\gamma}^2_{\mathrm{MM}} \to \gamma^{2\star}$ *in probability and* $\mathbb{E}_{\mathbb{P}^\star}\left[\left|\hat{\gamma}^2_{\mathrm{MM}} - \gamma^{2\star}\right|\right] \to 0$ *as* $K \to \infty$.

Finally, the maximum likelihood estimates $\hat{\mu}, \hat{\gamma}^2$ defined with the calibration studies are consistent.

Theorem 7. *Let* $\hat{\mu}, \hat{\gamma}^2$ *be the maximum likelihood estimates from the calibration marginal likelihood. Under the assumptions of Theorem 6,*

$$\|\hat{\mu}(\hat{\gamma}^2) - \mu^\star, \hat{\gamma}^2 - \gamma^{2\star}\|_2 = o_{\mathbb{P}^\star}(1). \tag{D.3}$$

Theorem 7 states that $(\hat{\mu}, \hat{\gamma}^2)$ is consistent in $\mathbb{P}^\star$-probability. While this does not imply convergence in $\mathbb{P}^\star$-expectation of the same quantity, a sufficient condition is *uniform integrability* of the error sequence. For example, if there exists $\delta > 0$ such that

$$\sup_{J \geq 1} \mathbb{E}_{\mathbb{P}^\star}\left[\|(\hat{\mu}, \hat{\gamma}^2) - (\mu^\star, \gamma^{2\star})\|_2^{1+\delta}\right] < \infty,$$

then (Billingsley, 2013, Theorem 3.5) implies $\mathbb{E}_{\mathbb{P}^\star}\left[\|(\hat{\mu}, \hat{\gamma}^2) - (\mu^\star, \gamma^{2\star})\|_2\right] \to 0$. More generally, any condition implying that $\{\|(\hat{\mu}, \hat{\gamma}^2) - (\mu^\star, \gamma^{2\star})\|_2\}_{J \geq 1}$ is uniformly integrable suffices to upgrade (D.3) from $o_{\mathbb{P}^\star}(1)$ to convergence in $L^1(\mathbb{P}^\star)$.

Appendix E: Proofs

Lemma 1. *The marginal likelihood objective in Eq. (3.19) has the form*

$$\begin{aligned}\log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}; \mu, \gamma^2) \overset{c}{=} \frac{1}{2}\Bigg[& \frac{\left(\sigma_{\mathrm{e}}^{-2} y_{\mathrm{e}} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \gamma^2)^{-1}(y_{\mathrm{o},j} - \mu)\right)^2}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \gamma^2)^{-1}} \\ & - \frac{y_{\mathrm{e}}^2}{\sigma_{\mathrm{e}}^2} - \sum_{j=1}^J \frac{(y_{\mathrm{o},j} - \mu)^2}{\sigma_{\mathrm{o},j}^2 + \gamma^2} - \sum_{j=1}^J \log(\sigma_{\mathrm{o},j}^2 + \gamma^2) \\ & - \log\left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \gamma^2)^{-1}\right)\Bigg].\end{aligned} \tag{E.1}$$

Proof of Lemma 1. After integrating out b_j, we have

$$y_{\mathrm{o},j} \mid \theta \sim \mathcal{N}(\theta + \mu, \sigma_{\mathrm{o},j}^2 + \gamma^2).$$

Hence

$$\begin{aligned}\log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}; \mu, \gamma^2) &= \log \mathrm{p}(y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}} \mid \theta; \mu, \gamma^2) - \log \mathrm{p}(\theta \mid y_{\mathrm{e}}, \mathbf{y}_{\mathrm{o}}; \mu, \gamma^2) \\ &\overset{c}{=} -\frac{(y_{\mathrm{e}} - \theta)^2}{2\sigma_{\mathrm{e}}^2} - \sum_{j=1}^J \frac{(y_{\mathrm{o},j} - \theta - \mu)^2}{2(\sigma_{\mathrm{o},j}^2 + \gamma^2)} \\ &\quad - \frac{1}{2}\sum_{j=1}^J \log(\sigma_{\mathrm{o},j}^2 + \gamma^2) + \frac{(\theta - \mu_{\mathrm{post}})^2}{2\sigma_{\mathrm{post}}^2} + \frac{1}{2}\log \sigma_{\mathrm{post}}^2.\end{aligned}$$

The posterior precision is

$$\sigma_{\text{post}}^{-2} = \sigma_{\text{e}}^{-2} + \sum_{j=1}^{J} (\sigma_{\text{o},j}^2 + \gamma^2)^{-1},$$

and the posterior mean is

$$\mu_{\text{post}} = \sigma_{\text{post}}^2 \left(\sigma_{\text{e}}^{-2} y_{\text{e}} + \sum_{j=1}^{J} (\sigma_{\text{o},j}^2 + \gamma^2)^{-1} (y_{\text{o},j} - \mu) \right).$$

Since the left-hand side does not depend on θ, we may collect only the terms free of θ:

$$\begin{aligned} \log \text{p}(y_{\text{e}}, \mathbf{y}_{\text{o}}; \mu, \gamma^2) \stackrel{c}{=} & -\frac{y_{\text{e}}^2}{2\sigma_{\text{e}}^2} - \frac{1}{2} \sum_{j=1}^{J} \frac{(y_{\text{o},j} - \mu)^2}{\sigma_{\text{o},j}^2 + \gamma^2} \\ & - \frac{1}{2} \sum_{j=1}^{J} \log(\sigma_{\text{o},j}^2 + \gamma^2) + \frac{\mu_{\text{post}}^2}{2\sigma_{\text{post}}^2} + \frac{1}{2} \log \sigma_{\text{post}}^2. \end{aligned}$$

Substituting the expressions for μ_{post} and σ_{post}^2 yields the result. □

Proof of Theorem 1. A flat prior on the biases is equivalent to taking $\gamma^2 \to \infty$. Then, by Proposition 1, as $\gamma^2 \to \infty$, we obtain the posterior

$$\theta \mid y_{\text{e}}, \mathbf{y}_{\text{o}} \sim \mathcal{N}(y_{\text{e}}, \sigma_{\text{e}}^2).$$

Thus the posterior mean and variance are $\hat{\theta}_{\text{flat}} = y_{\text{e}}$ and $\hat{v}_{\text{flat}}^2 = \sigma_{\text{e}}^2$.

Finally, under the data-generating model $y_{\text{e}} \sim \mathcal{N}(\theta^\star, \sigma_{\text{e}}^2)$, we have $\mathbb{E}\left[\hat{\theta}_{\text{flat}}\right] = \mathbb{E}\left[y_{\text{e}}\right] = \theta^\star$, so the estimator is unbiased, and

$$R(\hat{\theta}_{\text{flat}}, \theta^\star) = \mathbb{E}\left[(\hat{\theta}_{\text{flat}} - \theta^\star)^2\right] = \mathbb{E}\left[(y_{\text{e}} - \theta^\star)^2\right] = \sigma_{\text{e}}^2.$$

□

Proof of Theorem 3. For fixed γ^2, the map $\mu \mapsto \log \text{p}(y_{\text{e}}, y_{\text{o},1:J}; \mu, \gamma^2)$ is strictly concave. Its first-order condition gives

$$\hat{\mu}(\gamma^2) = \frac{\sum_{j=1}^{J} (\sigma_{\text{o},j}^2 + \gamma^2)^{-1} y_{\text{o},j}}{\sum_{j=1}^{J} (\sigma_{\text{o},j}^2 + \gamma^2)^{-1}} - y_{\text{e}} =: \tilde{y}_{\text{o}}(\gamma^2) - y_{\text{e}}. \tag{E.2}$$

Plugging $\hat{\mu}(\gamma^2)$ into the posterior mean from Proposition 1 gives

$$\mu_{\text{post}}\big(\hat{\mu}(\gamma^2)\big) = y_{\text{e}}.$$

Hence the posterior mean remains y_{e} for any choice of γ^2, including the empirical Bayes estimate $\hat{\gamma}^2$.

Substituting Eq. (E.2) into the marginal likelihood,

$$2\log \mathrm{p}(y_{\mathrm{e}}, y_{\mathrm{o},1:J}; \hat{\mu}(\gamma^2), \gamma^2) \stackrel{c}{=} -\sum_{j=1}^{J} \frac{(y_{\mathrm{o},j} - \tilde{y}_{\mathrm{o}})^2}{\sigma_{\mathrm{o},j}^2 + \gamma^2} - \sum_{j=1}^{J} \log(\sigma_{\mathrm{o},j}^2 + \gamma^2) - \log\left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \gamma^2)^{-1}\right).$$

As $\gamma^2 \to \infty$, this tends to $-\infty$, so a maximizer $\hat{\gamma}^2 \in [0, \infty)$ exists. Finally,

$$\frac{1}{\sigma_{\mathrm{post}}^2} = \sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1} > \sigma_{\mathrm{e}}^{-2},$$

and therefore $\sigma_{\mathrm{post}}^2 < \sigma_{\mathrm{e}}^2$. □

Proof of Theorem 5. Assume $\mu^\star = 0$. Then

$$y_{\mathrm{o},j} \mid \theta \overset{iid}{\sim} \mathcal{N}(\theta, \sigma_{\mathrm{o},j}^2 + \gamma^{2\star}).$$

By Proposition 1, the empirical Bayes posterior is $\mathcal{N}(\hat{\theta}_{\mathrm{EB}_0}, \hat{v}_{\mathrm{EB}_0}^2)$, where

$$\hat{\theta}_{\mathrm{EB}_0} = y_{\mathrm{e}} + \frac{\sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1} (y_{\mathrm{o},j} - y_{\mathrm{e}})}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}, \quad \hat{v}_{\mathrm{EB}_0}^2 = \left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}\right)^{-1}.$$

Let $\zeta_{\mathrm{e}} = y_{\mathrm{e}} - \theta^\star$ and $\zeta_j = y_{\mathrm{o},j} - \theta^\star$. Then

$$\hat{\theta}_{\mathrm{EB}_0} - \theta^\star = \zeta_{\mathrm{e}} + \frac{\sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1} (\zeta_j - \zeta_{\mathrm{e}})}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}.$$

Let $w_j := (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}$ and $D := \sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} w_j$. Then

$$\hat{\theta}_{\mathrm{EB}_0} - \theta^\star = \zeta_{\mathrm{e}} + D^{-1} \sum_{j=1}^{J} w_j (\zeta_j - \zeta_{\mathrm{e}}) = \left(1 - D^{-1} \sum_{j=1}^{J} w_j\right) \zeta_{\mathrm{e}} + D^{-1} \sum_{j=1}^{J} w_j \zeta_j = \frac{\sigma_{\mathrm{e}}^{-2}}{D} \zeta_{\mathrm{e}} + \frac{1}{D} \sum_{j=1}^{J} w_j \zeta_j.$$

Therefore, the risk evaluates to

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) = \mathbb{E}\left[\left(\frac{\sigma_{\mathrm{e}}^{-2}}{D} \zeta_{\mathrm{e}} + \frac{1}{D} \sum_{j=1}^{J} w_j \zeta_j\right)^2\right].$$

Since ζ_{e} is independent of $\zeta_1, \ldots, \zeta_J$ and all have mean zero, the cross term vanishes, so

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) = \mathbb{E}\left[\frac{\sigma_{\mathrm{e}}^{-4}}{D^2} \zeta_{\mathrm{e}}^2\right] + \mathbb{E}\left[\frac{1}{D^2} \left(\sum_{j=1}^{J} w_j \zeta_j\right)^2\right].$$

Now $\mathbb{E}\left[\zeta_{\mathrm{e}}^2\right] = \sigma_{\mathrm{e}}^2$, and since $\hat{\gamma}^2$ is independent of $\zeta_1, \cdots, \zeta_J$, we have

$$\mathbb{E}\left[\left(\sum_{j=1}^{J} w_j \zeta_j\right)^2 \middle| \hat{\gamma}^2\right] = \sum_{j=1}^{J} w_j^2(\sigma_{\mathrm{o},j}^2 + \gamma^{2\star}).$$

Hence,

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) = \mathbb{E}\left[\frac{\sigma_{\mathrm{e}}^{-2}}{D^2}\right] + \mathbb{E}\left[\frac{\sum_{j=1}^{J} w_j^2(\sigma_{\mathrm{o},j}^2 + \gamma^{2\star})}{D^2}\right].$$

Using

$$w_j^2(\sigma_{\mathrm{o},j}^2 + \gamma^{2\star}) = w_j^2(\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2) + w_j^2(\gamma^{2\star} - \hat{\gamma}^2) = w_j + w_j^2(\gamma^{2\star} - \hat{\gamma}^2),$$

we obtain

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) = \mathbb{E}\left[\frac{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} w_j}{D^2}\right] + \mathbb{E}\left[\frac{\sum_{j=1}^{J} w_j^2(\gamma^{2\star} - \hat{\gamma}^2)}{D^2}\right].$$

Since $D = \sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J} w_j$, the first term equals $\mathbb{E}\left[D^{-1}\right]$. For the second term, note that $0 \le \sum_{j=1}^{J} w_j^2/D^2 \le 1$, so

$$\left|\frac{\sum_{j=1}^{J} w_j^2(\gamma^{2\star} - \hat{\gamma}^2)}{D^2}\right| \le |\gamma^{2\star} - \hat{\gamma}^2|.$$

Therefore,

$$R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) \le \mathbb{E}\left[\frac{1}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J}(\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}\right] + \mathbb{E}\left[|\hat{\gamma}^2 - \gamma^{2\star}|\right].$$

It remains to show $\sum_{j=1}^{J}(\sigma_{\mathrm{o},j}^2 + \gamma^{2\star})^{-1} \to \infty$. If $\sigma_{\mathrm{o},j}^2 < \gamma^{2\star}$ infinitely often, the claim is immediate. Otherwise, for all sufficiently large j,

$$(\sigma_{\mathrm{o},j}^2 + \gamma^{2\star})^{-1} \ge \frac{1}{2}\sigma_{\mathrm{o},j}^{-2},$$

and the claim follows from $\sum_{j=1}^{J} \sigma_{\mathrm{o},j}^{-2} \to \infty$.

Since $\hat{\gamma}^2 \to \gamma^{2\star}$ in L_1, it also converges in probability. Thus, by the continuous mapping theorem,

$$\left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^{J}(\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}\right)^{-1} = o_{\mathbb{P}}(1).$$

Since this sequence is uniformly bounded by the interval $[0, \sigma_{\mathrm{e}}^2]$, the dominated convergence theorem implies that the expectation of the LHS above under $\mathbb{P}^\star$ also converges to 0.

Finally, combining this with the assumption that $\mathbb{E}\left[|\hat{\gamma}^2 - \gamma^{2\star}|\right] \to 0$ allows us to conclude that $R(\hat{\theta}_{\mathrm{EB}_0}, \theta^\star) \to 0$. □

Proof of Proposition 2. Let $v_j(\gamma^2) := \sigma_{\mathrm{o},j}^2 + \gamma^2$ and $v_j^\star := \sigma_{\mathrm{o},j}^2 + \gamma^{2\star}$. After integrating out θ, the log marginal likelihood is

$$\ell_J(\gamma^2) \overset{c}{=} -\frac{1}{2}\sum_{j=J+1}^{2J} \log v_j(\gamma^2) - \frac{1}{2}\log\left(\sum_{j=J+1}^{2J} v_j(\gamma^2)^{-1}\right) - \frac{1}{2}\sum_{j=J+1}^{2J} v_j(\gamma^2)^{-1}\big(y_{\mathrm{o},j} - \bar{y}_J(\gamma^2)\big)^2,$$

where

$$\bar{y}_J(\gamma^2) = \frac{\sum_{j=J+1}^{2J} v_j(\gamma^2)^{-1} y_{\mathrm{o},j}}{\sum_{j=J+1}^{2J} v_j(\gamma^2)^{-1}}.$$

Consider the criterion function

$$\tilde{\ell}_J(\gamma^2) := -\frac{1}{2}\sum_{j=J+1}^{2J} \underbrace{\left[\log v_j(\gamma^2) + \frac{(y_{\mathrm{o},j} - \theta^\star)^2}{v_j(\gamma^2)}\right]}_{=:g_j(\gamma^2)}.$$

Because $\sum_{j=J+1}^{2J} v_j(\gamma^2)^{-1} \to \infty$ uniformly on compact sets, $\bar{y}_J(\gamma^2) \to \theta^\star$ uniformly on compact sets in probability, and hence

$$\sup_{\gamma^2 \in S} |\ell_J(\gamma^2) - \tilde{\ell}_J(\gamma^2)| = o_{\mathbb{P}}(J), \tag{E.3}$$

for every compact $S \subset [0, \infty)$.

Now

$$\mathbb{E}\left[\tilde{\ell}_J(\gamma^2)\right] = -\frac{1}{2}\sum_{j=J+1}^{2J}\left[\log v_j(\gamma^2) + \frac{v_j^\star}{v_j(\gamma^2)}\right].$$

Therefore,

$$\mathbb{E}\left[\tilde{\ell}_J(\gamma^2) - \tilde{\ell}_J(\gamma^{2\star})\right] = -\frac{1}{2}\sum_{j=J+1}^{2J}\left[\log\frac{v_j(\gamma^2)}{v_j^\star} + \frac{v_j^\star}{v_j(\gamma^2)} - 1\right].$$

For each $\gamma^2 \in S \subset [0, B]$, recall that $\tilde{\ell}_J(\gamma^2) = \sum_{j=J+1}^{2J} g_j(\gamma^2)$. The functions $g_j(\gamma^2)$ are independent across j and continuous in γ^2. Since $\sup_{j\geq 1} \sigma_{\mathrm{o},j}^2 < \infty$ and $\gamma^2 \in [0, B]$, we have the uniform bound

$$\sup_{\gamma^2 \in S} |g_j(\gamma^2)| \leq C\big(1 + (y_{\mathrm{o},j} - \theta^\star)^2\big)$$

for a constant C depending only on B and $\sup_{j\geq 1} \sigma_{\mathrm{o},j}^2$. Moreover, the envelope has finite expectation as $\mathbb{E}\left[(y_{\mathrm{o},j} - \theta^\star)^2\right] < \infty$.

Moreover,

$$\frac{\partial g_j(\gamma^2)}{\partial \gamma^2} = -\frac{1}{2}\left[\frac{1}{\sigma_{\mathrm{o},j}^2 + \gamma^2} - \frac{(y_{\mathrm{o},j} - \theta^\star)^2}{(\sigma_{\mathrm{o},j}^2 + \gamma^2)^2}\right],$$

which is uniformly bounded in $L_1(\mathbb{P}_{\theta^\star,0,\gamma^{2\star}})$ over $\gamma^2 \in S$. Hence the class $\{g_j(\gamma^2) : \gamma^2 \in S\}$ is Lipschitz in γ^2 in the $L_1(\mathbb{P}_{\theta^\star,0,\gamma^{2\star}})$ with an integrable envelope, and

therefore forms a Glivenko–Cantelli class by (van der Vaart, 2000, Theorem 19.4).

Then, applying the Glivenko–Cantelli theorem (van der Vaart, 2000, Theorem 19.1) gives

$$\sup_{\gamma^2 \in S} \frac{1}{J} \left| \tilde{\ell}_J(\gamma^2) - \mathbb{E}\left[\tilde{\ell}_J(\gamma^2)\right] \right| \to 0 \qquad \text{in } [\mathbb{P}_{\theta^\star, 0, \gamma^{2\star}}]\text{-probability.}$$

Combining the above display with Eq. (E.3) gives

$$\sup_{\gamma^2 \in S} \frac{1}{J} \left| \ell_J(\gamma^2) - \mathbb{E}\left[\tilde{\ell}_J(\gamma^2)\right] \right| \to 0 \qquad \text{in } [\mathbb{P}_{\theta^\star, 0, \gamma^{2\star}}]\text{-probability.}$$

By the argmax theorem (van der Vaart and Wellner, 2023, Theorem 3.2.2), we conclude that $\hat{\gamma}^2 \to \gamma^{2\star}$ in probability.

Since $\hat{\gamma}^2 \le B$, the sequence of estimators $\{\hat{\gamma}^2\}$ is uniformly integrable. Thus convergence in probability implies convergence in L^1, so $\mathbb{E}\left[|\hat{\gamma}^2 - \gamma^{2\star}|\right] \to 0$. □

Proof of Proposition 3. The estimator $\hat{\gamma}^2_{\mathrm{MM}}$ is unbiased for $\gamma^{2\star}$. Moreover,

$$\mathrm{Var}\left[\hat{\gamma}^2_{\mathrm{MM}}\right] = \frac{2}{J^2} \sum_{j=J+1}^{2J} \sigma^4_{\mathrm{o},j} \to 0.$$

Hence $\mathbb{E}\left[\left|\hat{\gamma}^2_{\mathrm{MM}} - \gamma^{2\star}\right|^2\right] \to 0$, thus $\mathbb{E}\left[\left|\hat{\gamma}^2_{\mathrm{MM}} - \gamma^{2\star}\right|\right] \to 0$ by Hölder's inequality. □

Proof of Theorem 6. Without loss of generality, assume $\sigma^2_{\mathrm{c},1} \le \sigma^2_{\mathrm{c},2} \le \cdots$. A direct expansion gives

$$R(\hat{\theta}_{\mathrm{CEB}}, \theta^\star) = \sigma^2_{\mathrm{e}} - 2A_1 + A_2 + A_3,$$

where

$$A_1 = \sigma^2_{\mathrm{e}} \mathbb{E}\left[\frac{\sum_{j=1}^J (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1}}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1}}\right],$$

$$A_2 = \mathbb{E}\left[\left(\frac{\sum_{j=1}^J (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1}(\mu^\star - \hat{\mu})}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1}}\right)^2\right],$$

$$A_3 = \mathbb{E}\left[\frac{\sum_{j=1}^J (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-2}(\sigma^2_{\mathrm{o},j} + \gamma^{2\star} + \sigma^2_{\mathrm{e}})}{\left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma^2_{\mathrm{o},j} + \hat{\gamma}^2)^{-1}\right)^2}\right].$$

To control A_2, note that

$$\frac{(\gamma^2 + \sigma^2_{\mathrm{c},k})^{-1}}{\sum_{j=1}^K (\gamma^2 + \sigma^2_{\mathrm{c},j})^{-1}} \le \frac{\sigma^{-2}_{\mathrm{c},1}}{\sum_{j=1}^K \sigma^{-2}_{\mathrm{c},j}}. \tag{E.4}$$

Thus

$$A_2 \le \mathbb{E}\left[|\hat{\mu}(\hat{\gamma}^2) - \mu^\star|^2\right] = \mathbb{E}\left[\mathrm{Var}\left[\hat{\mu}(\hat{\gamma}^2) \mid \hat{\gamma}^2\right]\right] \le \sup_{\gamma^2 \ge 0} \frac{\sum_{k=1}^K (\gamma^2 + \sigma_{\mathrm{c},k}^2)^{-2}(\gamma^{2\star} + \sigma_{\mathrm{c},k}^2)}{\left(\sum_{k=1}^K (\gamma^2 + \sigma_{\mathrm{c},k}^2)^{-1}\right)^2}.$$

For A_3, write

$$\begin{aligned} A_3 &= \mathbb{E}\left[\frac{\sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}{\left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}\right)^2}\right] + A_1 \\ &\quad + \mathbb{E}\left[\frac{\sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-2}(\gamma^{2\star} - \hat{\gamma}^2)}{\left(\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}\right)^2}\right] \\ &\le \mathbb{E}\left[\frac{1}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}\right] + \mathbb{E}\left[|\gamma^{2\star} - \hat{\gamma}^2|\right] + A_1. \end{aligned}$$

The first term tends to 0 and the second tends to 0 by assumption, so $A_3 \to A_1$.

Therefore

$$R(\hat{\theta}_{\mathrm{CEB}}, \theta^\star) = \sigma_{\mathrm{e}}^2 - 2A_1 + A_3 = \sigma_{\mathrm{e}}^2 - A_1 + o(1) = \mathbb{E}\left[\frac{\sigma_{\mathrm{e}}^2}{\sigma_{\mathrm{e}}^{-2} + \sum_{j=1}^J (\sigma_{\mathrm{o},j}^2 + \hat{\gamma}^2)^{-1}}\right] + o(1) \to 0.$$

□

Proof of Proposition 4. Under the calibration model,

$$y_{\mathrm{c},k} = \mu^\star + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, \gamma^{2\star} + \sigma_{\mathrm{c},k}^2),$$

where $\varepsilon_1, \ldots, \varepsilon_K$ are independent. Write $v_k := \gamma^{2\star} + \sigma_{\mathrm{c},k}^2$. Then

$$y_{\mathrm{c},k} - \bar{y}_{\mathrm{c}} = \varepsilon_k - \bar{\varepsilon}, \qquad \bar{\varepsilon} = \frac{1}{K}\sum_{\ell=1}^K \varepsilon_\ell.$$

Hence

$$\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star} = \frac{1}{K}\sum_{k=1}^K \left((\varepsilon_k - \bar{\varepsilon})^2 - v_k\right).$$

We first compute the bias. Since

$$\mathbb{E}\left[(\varepsilon_k - \bar{\varepsilon})^2\right] = v_k + \mathrm{Var}\left[\bar{\varepsilon}\right] - 2\mathrm{Cov}(\varepsilon_k, \bar{\varepsilon}) = v_k + \frac{1}{K^2}\sum_{\ell=1}^K v_\ell - \frac{2}{K} v_k,$$

we obtain

$$\mathbb{E}\left[\hat{\gamma}_{\mathrm{MM}}^2\right] = \frac{1}{K}\sum_{k=1}^K \left(\mathbb{E}\left[(\varepsilon_k - \bar{\varepsilon})^2\right] - \sigma_{\mathrm{c},k}^2\right).$$

Using $v_k = \gamma^{2\star} + \sigma_{\mathrm{c},k}^2$ gives

$$\mathbb{E}\left[\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star}\right] = -\frac{1}{K^2}\sum_{k=1}^{K} v_k = -\frac{1}{K^2}\sum_{k=1}^{K}(\gamma^{2\star} + \sigma_{\mathrm{c},k}^2).$$

The assumption $\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^4/K^2 \to 0$ implies $\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^2/K^2 \to 0$. Indeed, by Cauchy–Schwarz,

$$\left(\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^2\right)^2 \le K \sum_{k=1}^{K} \sigma_{\mathrm{c},k}^4.$$

Taking square roots and dividing by K^2 yields

$$\frac{1}{K^2}\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^2 \le \sqrt{\frac{1}{K^3}\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^4} = \sqrt{\frac{1}{K} \cdot \frac{1}{K^2}\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^4} \to 0.$$

Hence $\mathbb{E}\left[\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star}\right] \to 0$.

Next we control the variance. Since $\hat{\gamma}_{\mathrm{MM}}^2$ is a quadratic form of the independent Gaussian variables $\varepsilon_1, \ldots, \varepsilon_K$, a standard calculation gives

$$\mathrm{Var}\left[\hat{\gamma}_{\mathrm{MM}}^2\right] \lesssim \frac{1}{K^2}\sum_{k=1}^{K} v_k^2 = \frac{1}{K^2}\sum_{k=1}^{K}(\gamma^{2\star} + \sigma_{\mathrm{c},k}^2)^2.$$

Since $\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^4/K^2 \to 0$, it follows that $\mathrm{Var}\left[\hat{\gamma}_{\mathrm{MM}}^2\right] \to 0$.

Combining the bias and variance bounds,

$$\mathbb{E}\left[(\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star})^2\right] = \mathrm{Var}\left[\hat{\gamma}_{\mathrm{MM}}^2\right] + \left(\mathbb{E}\left[\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star}\right]\right)^2 \to 0.$$

Finally, by Cauchy–Schwarz,

$$\mathbb{E}\left[|\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star}|\right] \le \sqrt{\mathbb{E}\left[(\hat{\gamma}_{\mathrm{MM}}^2 - \gamma^{2\star})^2\right]} \to 0.$$

Therefore, the moment estimator is consistent in the L_1 sense. □

Lemma 2. *Let S be a compact set in $[0, \infty)$. Under the assumptions of Theorem 6, the profile MLE $\hat{\mu}(\gamma^2)$ satisfies*

$$\sup_{\gamma^2 \in S} |\hat{\mu}(\gamma^2) - \mu^\star| = o_{\mathbb{P}^\star}(1). \tag{E.5}$$

Proof of Lemma 2. The profile MLE is

$$\hat{\mu}(\gamma^2) = \frac{\sum_{k=1}^{K}(\gamma^2 + \sigma_{\mathrm{c},k}^2)^{-1} y_{\mathrm{c},k}}{\sum_{k=1}^{K}(\gamma^2 + \sigma_{\mathrm{c},k}^2)^{-1}}.$$

Write $y_{c,k} = \mu^\star + \sqrt{\gamma^{2\star} + \sigma_{c,k}^2}\, z_k$ with $z_k \overset{iid}{\sim} \mathcal{N}(0,1)$. Then

$$\hat{\mu}(\gamma^2) - \mu^\star = \frac{\sum_{k=1}^K (\gamma^2 + \sigma_{c,k}^2)^{-1} \sqrt{\gamma^{2\star} + \sigma_{c,k}^2}\, z_k}{\sum_{k=1}^K (\gamma^2 + \sigma_{c,k}^2)^{-1}}.$$

Let $B = \max(S)$ and define

$$X(\gamma^2) := \sum_{k=1}^K (\gamma^2 + \sigma_{c,k}^2)^{-1} \sqrt{\gamma^{2\star} + \sigma_{c,k}^2}\, z_k.$$

Then

$$\sup_{\gamma^2 \in S} |\hat{\mu}(\gamma^2) - \mu^\star| \le \frac{\sup_{\gamma^2 \in S} |X(\gamma^2)|}{\sum_{k=1}^K \sigma_{c,k}^{-2}}. \tag{E.6}$$

The process $X(\gamma^2)$ is Gaussian with pointwise variance bounded by

$$V_{\max} \le \sum_{k=1}^K \frac{\gamma^{2\star} + \sigma_{c,k}^2}{\sigma_{c,k}^4}.$$

Its canonical metric satisfies

$$d(s,t) \le L|s-t|, \qquad L := \sqrt{\sum_{k=1}^K \sigma_{c,k}^{-8} \gamma^{2\star} + \sum_{k=1}^K \sigma_{c,k}^{-6}}.$$

Since $S \subset \mathbb{R}$ is one-dimensional, standard Gaussian process bounds give

$$\sup_{\gamma^2 \in S} |X(\gamma^2)| = O_{\mathbb{P}^\star}\left(BL\sqrt{\log K}\right).$$

Combining with Eq. (E.6),

$$\sup_{\gamma^2 \in S} |\hat{\mu}(\gamma^2) - \mu^\star| = O_{\mathbb{P}^\star}\left(\frac{B\sqrt{\log K \left(\sum_{k=1}^K \sigma_{c,k}^{-8} \gamma^{2\star} + \sum_{k=1}^K \sigma_{c,k}^{-6}\right)}}{\sum_{k=1}^K \sigma_{c,k}^{-2}}\right).$$

Under the assumptions of Theorem 6, the right-hand side is $o_{\mathbb{P}^\star}(1)$. □

Proof of Theorem 7. The calibration log-likelihood is

$$\log \mathrm{p}(\boldsymbol{y}_c; \mu, \gamma^2) = -\frac{1}{2} \sum_{k=1}^K \left[\frac{(y_{c,k} - \mu)^2}{\gamma^2 + \sigma_{c,k}^2} + \log(\gamma^2 + \sigma_{c,k}^2)\right].$$

For fixed γ^2, the profiled MLE is

$$\hat{\mu}(\gamma^2) = \frac{\sum_{k=1}^K (\gamma^2 + \sigma_{c,k}^2)^{-1} y_{c,k}}{\sum_{k=1}^K (\gamma^2 + \sigma_{c,k}^2)^{-1}}.$$

Substituting into the score for γ^2 yields

$$\tilde{\psi}_K(\gamma^2) = \frac{1}{2}\sum_{k=1}^{K}\left[\frac{(y_{c,k} - \hat{\mu}(\gamma^2))^2}{(\gamma^2 + \sigma_{c,k}^2)^2} - \frac{1}{\gamma^2 + \sigma_{c,k}^2}\right].$$

Write $y_{c,k} = \mu^\star + \sqrt{\gamma^{2\star} + \sigma_{c,k}^2}\, z_k$ with $z_k \overset{iid}{\sim} \mathcal{N}(0,1)$. By Lemma 2,

$$\sup_{\gamma^2 \in S} |\hat{\mu}(\gamma^2) - \mu^\star| = o_{\mathbb{P}^\star}(1).$$

Therefore,

$$\tilde{\psi}_K(\gamma^{2\star}) = O_{\mathbb{P}^\star}\left(\sqrt{\sum_{k=1}^{K}(\gamma^{2\star} + \sigma_{c,k}^2)^{-2}}\right) + o_{\mathbb{P}^\star}\left(\sum_{k=1}^{K}(\gamma^{2\star} + \sigma_{c,k}^2)^{-2}\right).$$

Define

$$f_K(\gamma^2) := \frac{1}{2}\sum_{k=1}^{K}\frac{\gamma^{2\star} - \gamma^2}{(\gamma^2 + \sigma_{c,k}^2)^2}.$$

Then $f_K(\gamma^{2\star}) = 0$ and $\gamma^{2\star}$ is a separated root. More precisely, for any $\delta > 0$,

$$\inf_{\substack{|\gamma^2 - \gamma^{2\star}| > \delta \\ \gamma^2 \in S}} |f_K(\gamma^2) - f_K(\gamma^{2\star})| \gtrsim \delta \sum_{k=1}^{K}(B \vee (\gamma^{2\star} + \sigma_{c,k}^2))^{-2}, \tag{E.7}$$

where $B = \max(S)$.

Since

$$\sup_{\gamma^2 \in S} |\tilde{\psi}_K(\gamma^2) - f_K(\gamma^2)| = o_{\mathbb{P}^\star}\left(\sum_{k=1}^{K}\sigma_{c,k}^{-4}\right),$$

for any sequence $M_K \to \infty$, we apply the triangle inequality and bound

$$\begin{aligned}
\mathbb{P}^\star\left(\left|\hat{\gamma}^2 - \gamma^{2\star}\right| > M_K\delta\right) &\le \mathbb{P}^\star\left(\left|f_K(\hat{\gamma}^2)\right| > 0.5\delta M_K r_K\right) \\
&\le \mathbb{P}^\star\left(\left|f_K(\hat{\gamma}^2) - \tilde{\psi}_K(\hat{\gamma}^2)\right| + \left|\tilde{\psi}_K(\hat{\gamma}^2)\right| > 0.5\delta M_K r_K\right) \\
&\le \mathbb{P}^\star\left(\sup_{\gamma^2 \in S}\left|f_K(\gamma^2) - \tilde{\psi}_K(\gamma^2)\right| + \left|\tilde{\psi}_K(\hat{\gamma}^2)\right| > 0.5\delta M_K r_K\right) \\
&= \mathbb{P}^\star\left(o_{\mathbb{P}^\star}\left(\sum_{k=1}^{K}\sigma_{c,k}^{-4}\right) > 0.5\delta M_K r_K\right) \to 0.
\end{aligned}$$

Hence, $\hat{\gamma}^2 \to \gamma^{2\star}$ in probability.

Combining this with Lemma 2, which gives

$$\sup_{\gamma^2 \in S} |\hat{\mu}(\gamma^2) - \mu^\star| = o_{\mathbb{P}^\star}(1),$$

we obtain

$$|\hat{\mu}(\hat{\gamma}^2) - \mu^\star| \leq \sup_{\gamma^2 \in S} |\hat{\mu}(\gamma^2) - \mu^\star| = o_{\mathbb{P}^\star}(1).$$

Therefore both coordinates converge to zero in probability, and thus

$$\|\hat{\mu}(\hat{\gamma}^2) - \mu^\star,\, \hat{\gamma}^2 - \gamma^{2\star}\|_2 = o_{\mathbb{P}^\star}(1).$$

Since

$$\sup_{\gamma^2 \in S} |\tilde{\psi}_K(\gamma^2) - f_K(\gamma^2)| = o_{\mathbb{P}^\star}\left(\sum_{k=1}^{K} \sigma_{\mathrm{c},k}^{-4}\right),$$

the usual estimating-equation argument implies $\hat{\gamma}^2 \to \gamma^{2\star}$ in probability. Combining this with Lemma 2 yields

$$\|\hat{\mu}(\hat{\gamma}^2) - \mu^\star,\, \hat{\gamma}^2 - \gamma^{2\star}\|_2 = o_{\mathbb{P}^\star}(1).$$

□

Appendix F: Details of the Real Data Study

In the real data study, we choose the outcome to be water usage for each household in summer 2008. The treatment is Treatment 3: the tip sheet plus a personalized letter with a social-comparison component, which shows each household how its past water use compares with the median county usage (a strong social-norm message). The control group did not receive any message.

Following the field experiment of Ferraro and Miranda (2013), we choose the pre-treatment covariates to include: (i) water usage from June to November 2006 (high—above median, or low—below median); (ii) fair market value of the house in the year the treatment was assigned; (iii) ownership status (whether the household owns or rents the house); (iv) age of the home; (v) education (whether the homeowner has a bachelor's degree); and (vi) race, measured as the percentage of white residents.

To construct the observational design, we specify a new propensity score model $e_{\mathrm{o}}(X) := \Pr_{\mathrm{o}}(A = 1 \mid X)$ in which treatment assignment depends on pre-treatment water consumption. In particular, we let $e_{\mathrm{o}}(X_i) = \sigma(\beta\, X_{i,\mathrm{summer2006}})$, where $\sigma(u) = (1 + e^{-u})^{-1}$ is the logistic link and $X_{i,\mathrm{summer2006}}$ denotes standardized water usage in summer 2006. We fix $\beta = 0.5$, so that households with higher historical water consumption are much more likely to receive the treatment.

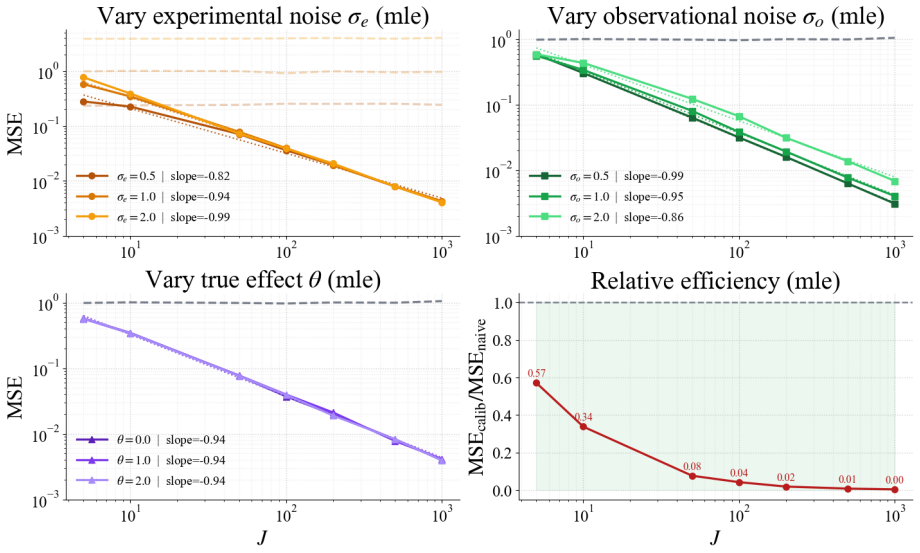


FIG F.1. *Mean squared error (MSE) of the calibrated empirical Bayes estimator* $\hat{\theta}_{\mathrm{CEB}}$ *versus sample size* J*, where the empirical Bayes estimates are based on maximum marginal likelihood.*